



Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

WaveNO: Solving PDEs using Physics-Informed Attention Bias

Master Thesis

Mamoun Chami

April 1, 2026

Advisors: Prof. Alexandre Bayen, Prof. Hossein Matin, Prof. Niao He

Department of Computer Science, ETH Zürich

Acknowledgments

I would like to express my sincere gratitude to all those who supported me throughout this thesis.

First and foremost, I am deeply grateful to Prof. Hossein Nick Zinat Matin, my co-supervisor at École Polytechnique, for his continuous guidance, insightful discussions, and unwavering support. His mentorship was instrumental in shaping this work.

I would like to thank my supervisors, Prof. Alexandre Bayen at UC Berkeley and Prof. Niao He at ETH Zürich, for giving me the opportunity to pursue this research and for their valuable feedback throughout the project.

I also thank Cassie Jeng, PhD student at UC Berkeley, and Katie Wang, undergraduate student at UC Berkeley, who contributed to the project.

Abstract

Neural operators have emerged as a promising paradigm for learning PDE solution operators, yet hyperbolic conservation laws remain a persistent challenge: solutions develop shock discontinuities in finite time, which standard architectures smear due to spectral bias, and recent theoretical results show that operators relying on linear reconstruction fundamentally cannot approximate these solution maps efficiently. We introduce WaveNO, a transformer-based neural operator that addresses these difficulties by embedding the physics of wave propagation directly into the architecture. WaveNO operates on exact piecewise constant initial conditions, breakpoint positions and segment values, rather than discretized grids, eliminating discretization-induced errors that can shift shock trajectories across the entire time horizon. The core mechanism is a physics-informed attention bias that encodes the causal structure of the PDE: for each space-time query point, the bias is derived from local Riemann problems and penalizes attention to segments outside their analytically determined influence zones. Feature-wise Linear Modulation conditions segment embeddings on time, enabling parallel prediction of the full space-time solution without autoregressive rollout. All physics features are supplied through a pluggable flux interface, making WaveNO equation-agnostic. We evaluate on the LWR traffic flow model and the ARZ system, demonstrating improved accuracy over FNO and DeepONet baselines on both in-distribution and out-of-distribution initial conditions, as well as robust generalization to higher output resolutions unseen during training.

Contents

Contents	iii
1 Introduction	1
2 Related Work	4
2.1 Numerical Methods for Conservation Laws	4
2.2 Neural Operators	4
2.3 Transformer-based Neural Operators	5
2.4 Physics-Informed Neural Networks	6
2.5 Neural Networks for Hyperbolic PDEs	7
3 Problem Formulation	9
3.1 Scalar Conservation Laws	9
3.1.1 The Rankine–Hugoniot Condition	9
3.1.2 The Riemann Problem	10
3.1.3 The LWR Traffic Flow Model	10
3.2 Systems of Conservation Laws	11
3.2.1 The ARZ Model	12
3.3 Operator Learning from Piecewise Constant Initial Conditions	12
4 Architecture	14
4.1 Segment Encoding	14
4.1.1 Segment Self-Attention	16
4.2 Query Encoding	17
4.3 FiLM Time Conditioning	17
4.4 Physics-Informed Attention Bias	18
4.4.1 Motivation and Relation to ALiBi	18
4.4.2 Wave Classification and Boundary Speeds	19
4.4.3 Similarity Variable Formulation	19
4.4.4 Extension to the ARZ System	21

4.5	Biased Cross-Attention	23
4.6	Density Head	23
4.7	Adaptability to Other PDEs	24
5	Experiments	25
5.1	Experimental Setup	25
5.1.1	Equations and Data Generation	25
5.1.2	Models and Ablations	26
5.1.3	Training and Evaluation Protocol	26
5.2	Overall Accuracy	27
5.3	Ablation: Component Contributions	27
5.3.1	Architecture Progression	28
5.3.2	Temporal Error Analysis	29
5.4	Generalization	30
5.4.1	Resolution Generalization	31
5.4.2	Initial Condition Complexity	32
5.5	Qualitative Analysis	33
5.6	Computational Efficiency	34
6	Discussion	36
6.1	Architectural Design Choices	36
6.1.1	Shared Bias vs. Per-Head Slopes	36
6.1.2	Linear Query Encoding vs. Fourier Features	37
6.1.3	Why No Self-Attention Among Queries	37
6.1.4	Implicit $1/t$ Scaling vs. Explicit Collision-Time Damping	38
6.2	Interpreting the Experimental Results	38
6.2.1	Outperforming a Numerical Solver	38
6.2.2	Resolution Generalization and Ground-Truth Quality	39
6.2.3	The Bias Override Problem	40
7	Conclusion	41
7.1	Summary	41
7.2	Limitations	41
7.3	Future Work	42
	Bibliography	43
	Appendix	51
	Experiment Details	51
	Training Hyperparameters	51
	Per-Equation Configuration	51
	Model Architectures	51
	Additional Experimental Results	53
	Full Solution Comparison	53

Chapter 1

Introduction

Partial differential equations (PDEs) govern the spatial and temporal evolution of physical quantities and arise across virtually all quantitative sciences and engineering disciplines. Among the principal classes of PDEs, hyperbolic equations are distinguished by finite propagation speeds and well-defined wavefronts [15]. They constitute the governing equations for a broad range of wave propagation phenomena, including traffic flow, gas dynamics and shallow water flows. A unifying mathematical framework for these problems is provided by conservation laws, in which the dynamics are determined by flux functions that dictate the speed and direction of information propagation [36]. The finite propagation speed distinguishes hyperbolic PDEs from parabolic equations (such as the heat equation, where disturbances diffuse instantaneously) and makes them both physically richer and computationally more challenging. In all of these applications, the practical need to solve the equations quickly and accurately for many different scenarios, varying initial conditions, boundary data, or physical parameters motivates the search for efficient computational methods.

A defining feature of hyperbolic conservation laws is that their solutions can develop discontinuities, shock waves and contact discontinuities, in finite time, even from arbitrarily smooth initial data [35, 11]. This fundamentally complicates numerical approximation. Classical solutions cease to exist after shock formation, requiring the notion of weak solutions, which in turn are not unique: entropy conditions must be imposed to select the physically relevant one [35, 51]. The Godunov scheme [17] established the modern paradigm of finite volume methods by solving local Riemann problems at cell interfaces, but as a first-order method it introduces excessive numerical diffusion that smears sharp fronts. Higher-order finite volume methods [36, 67] and discontinuous Galerkin schemes [9, 22] achieve better accuracy in smooth regions, but require sophisticated limiters or shock-capturing strategies to control spurious oscillations near discontinuities [16].

Crucially, resolving the thin layers around shocks demands locally refined meshes, making simulations computationally expensive, particularly for parametric studies that require solving the equations repeatedly for many different initial conditions.

The computational cost of classical solvers has motivated the development of neural operators: deep learning architectures that learn mappings between function spaces and can approximate PDE solution operators directly from data [31]. The Fourier Neural Operator (FNO) [41] and DeepONet [48] have achieved remarkable accuracy on a range of PDE benchmarks, while physics-informed neural networks (PINNs) [59] incorporate governing equations as soft constraints [28]. However, hyperbolic PDEs with discontinuous solutions remain a major open challenge. Lanthaler et al. [34] proved that neural operators relying on linear reconstruction, including standard DeepONet, fundamentally cannot efficiently approximate solution operators of conservation laws that develop shocks. In practice, neural networks exhibit a well-documented spectral bias toward low-frequency functions [56], causing predicted solutions to smear discontinuities into smooth transitions. Physics-informed losses fare no better: the PDE residual is undefined at shocks, and enforcing conservation in strong form across the domain leads to training instabilities. These theoretical and practical barriers indicate that standard architectures lack the structural inductive biases needed to capture the discontinuous, causally structured solutions of hyperbolic PDEs.

In this thesis, we introduce **WaveNO**, a transformer-based neural operator designed to address these challenges by embedding the physics of wave propagation directly into the network architecture. The central idea is that information in hyperbolic PDEs propagates along characteristics, curves in the space-time plane whose slopes are determined by the flux function, and this causal structure can be encoded as an inductive bias in the attention mechanism [71]. Specifically, WaveNO augments cross-attention between input segments and space-time query points with a characteristic attention bias: for each query point (t, x) , the bias modulates attention weights according to whether the point lies along a characteristic emanating from each input segment, encouraging the network to respect the physical information flow. Additionally, WaveNO employs Feature-wise Linear Modulation (FiLM) [53] to condition the decoder on time, enabling the network to capture the evolving dynamics of wavefronts. A crucial design choice is the representation of initial conditions: rather than feeding the model a discretized grid, WaveNO takes the exact piecewise constant description, breakpoint positions and segment values, as input. This is essential because hyperbolic conservation laws are highly sensitive to initial data: small perturbations or discretization errors in the initial conditions can lead to qualitatively different solution structures, with shocks appearing at different locations or entirely different wave interaction patterns emerging. By preserving the exact discontinuity

locations, WaveNO avoids these ambiguities and also enables mesh-invariant generalization, since the input representation is independent of any particular spatial discretization.

The main contributions of this thesis are:

1. We propose WaveNO, a mesh-invariant neural operator that maps initial conditions to full space-time solutions. By operating on the exact piecewise constant description of the initial data and querying arbitrary space-time coordinates, WaveNO is independent of any fixed spatial or temporal discretization.
2. We incorporate physics-informed features at two levels of the architecture: in the input, through characteristic-speed and flux-derived features that encode the local wave structure of the conservation law; and inside the model itself, through an attention bias that steers the cross-attention mechanism along characteristic directions, embedding the causal propagation structure of the PDE directly into the network.
3. Using these components, we reduce test MSE by **63%** over FNO on the LWR traffic flow equation and by **30%** on the ARZ system, while also outperforming a first-order Godunov solver evaluated on the same grid.

Code to reproduce all experiments is available at <https://github.com/mchami02/waveno>.

Related Work

2.1 Numerical Methods for Conservation Laws

The numerical solution of hyperbolic conservation laws has been an active area of research since the mid-twentieth century. The Godunov scheme [17] introduced the paradigm of solving Riemann problems at cell interfaces, later made practical by approximate solvers such as Roe’s linearization [63]. Higher-order accuracy was achieved through reconstruction-based methods: van Leer’s MUSCL scheme [70] uses piecewise linear reconstruction with slope limiters, while ENO [21] and WENO [45, 64] schemes adaptively select or blend stencils to achieve high-order accuracy without spurious oscillations. Discontinuous Galerkin methods [9, 22] offer an alternative by combining finite element flexibility with finite volume locality, though they still require limiters or artificial viscosity near shocks. Adaptive mesh refinement [4] concentrates resolution near discontinuities, but adds implementation complexity. Comprehensive treatments can be found in LeVeque [36], Toro [67], and Godlewski and Raviart [16]. Despite decades of progress, all these methods share a fundamental cost: resolving shocks on a grid requires fine meshes, making parametric studies over many initial conditions computationally expensive and motivating data-driven alternatives.

2.2 Neural Operators

Neural operators learn mappings between infinite-dimensional function spaces, enabling a single trained model to approximate the solution operator of a PDE for any input function within a given class [31]. Two dominant architectural paradigms have emerged in the literature.

The first paradigm adopts an encoder-decoder structure: one network encodes the input function into a latent representation, and a second network decodes the output at each query location independently. This design origi-

nates from the universal approximation theorem for operators due to Chen and Chen [7], which guarantees that shallow networks can approximate any nonlinear continuous operator. DeepONet [48] realizes this through a branch–trunk decomposition, where the branch encodes the input function and the trunk encodes the query location, with their dot product producing the prediction. Lanthaler et al. [33] established rigorous error bounds for this architecture. Subsequent extensions preserve the same encoder–decoder principle: POD-DeepONet [49] replaces the trunk with pre-computed POD bases, and MIONet [27] generalizes the branch–trunk product to multiple input functions.

The second paradigm processes the entire discretized field jointly, maintaining spatial relationships between outputs throughout the network. The Fourier Neural Operator (FNO) [41], building on the graph-based kernel parameterization of Li et al. [40], applies global convolutions in Fourier space via the fast Fourier transform, so that each output point is coupled to every other point at every layer. Extensions such as Geo-FNO [39], Factorized FNO [68], and U-NO [58] retain this grid-to-grid structure while addressing limitations in geometry handling, depth, and multi-scale resolution.

Both paradigms, however, encounter fundamental difficulties when the target solution contains shocks. FNO’s truncated Fourier basis is inherently global and smooth, causing it to oscillate near discontinuities (especially in high resolutions), a manifestation of the spectral bias common to frequency-domain representations. DeepONet’s limitation is more structural: Lanthaler et al. [34] proved that any method relying on a linear reconstruction step, including the standard branch–trunk dot product, cannot efficiently approximate solution operators of PDEs with discontinuities, establishing that nonlinear reconstruction mechanisms are necessary.

The Wavelet Neural Operator (WNO) [69] attempts to mitigate the spectral smoothing by replacing the Fourier kernel with a wavelet-based one that localizes in both space and frequency, offering better resolution of sharp features. However, it does not embed equation-specific structure such as characteristic speeds or shock relations into the operator, which is the direction pursued in this work.

2.3 Transformer-based Neural Operators

The transformer architecture [71], originally developed for natural language processing, has been increasingly applied to PDE operator learning due to its ability to model long-range dependencies through attention. Cao [6] showed that linear attention (without softmax) can be interpreted as a kernel integral transform with approximation capacity comparable to a Petrov–Galerkin projection, and Calvello et al. [5] later formalized this connection rigorously,

establishing approximation-theoretic foundations for continuous attention as a map between function spaces.

A central design pattern is cross-attention between input observations and output query coordinates, which decouples the model from any fixed discretization. OFormer [37], GNOT [20], and LOCA [30] established this approach, differing mainly in how they normalize attention and encode query locations. CViT [74] showed that the choice of coordinate embedding is critical for capturing multi-scale features and mitigating spectral bias, and ViTO [52] extended the paradigm to inverse problems, reconstructing source terms or initial conditions from coarse observations. Scaling to large grids has been addressed through axial factorization [38], physics-aware domain partitioning [77], and mesh-decoupled encode-process-decode frameworks [1]. A separate line of work explores pretraining across multiple PDE families [57, 19, 50], learning shared representations that transfer to unseen equations.

Wu et al. [76] (FlashBias) showed that attention with additive spatial distance biases can be computed efficiently, and demonstrated that such biases improve accuracy in PDE solving tasks, confirming that distance-aware attention is beneficial in this domain. However, all of the above methods employ generic attention: the attention weights are learned from data or modulated by Euclidean distance, without incorporating equation-specific structure such as characteristic speeds or shock relations. This is the gap that the present work addresses.

2.4 Physics-Informed Neural Networks

Physics-informed neural networks (PINNs) [59] embed the governing PDE directly into the training loss by minimizing the residual at collocation points, allowing forward and inverse problems to be solved without labeled data beyond boundary and initial conditions. Karniadakis et al. [28] survey this paradigm and its extensions. The appeal is clear, the PDE provides a dense, data-free supervisory signal, but two independent lines of analysis have revealed that this signal creates pathological optimization landscapes. Wang et al. [75] showed through a neural tangent kernel analysis that PDE residual, boundary, and initial-condition loss components converge at vastly different rates, causing training to stall. Krishnapriyan et al. [32] demonstrated that PINNs fail even on simple convection-dominated problems, tracing the failure to the loss landscape itself rather than to limited network expressivity. Wang et al. [73] provide a comprehensive practical guide to mitigating these issues. De Ryck and Mishra [12] complemented these empirical findings with rigorous approximation-error bounds for PINNs, DeepONets, and FNOs, showing in particular that physics-informed learning can mitigate the curse

of dimensionality for nonlinear parabolic PDEs.

Two broad strategies have emerged to address these training difficulties. The first adapts the loss formulation. Domain-decomposition approaches, conservative PINNs (cPINNs) [26] and extended PINNs (XPINNs) [25], partition the domain into sub-problems with separately trained networks, enforcing flux continuity at interfaces in a manner analogous to finite volume methods. Variational formulations such as hp-VPINNs [29] replace the pointwise residual with a weak form using piecewise polynomial test functions, reducing the differential order of the loss and improving accuracy for rough solutions. For hyperbolic conservation laws specifically, wPINNs [13] substitute the pointwise residual, which is undefined at shocks, with a weak formulation based on Kruřkov entropy conditions, recasting training as a min-max problem with provable L^1 error bounds. The second strategy adapts the architecture: Coutinho et al. [10] learned spatially varying artificial viscosity maps, localizing diffusion near discontinuities rather than relying on manual tuning. A hybrid approach is the Physics-Informed Neural Operator (PINO) [42], which combines supervised training on low-resolution data with a PDE residual loss evaluated at higher resolution, enabling zero-shot super-resolution beyond the training grid.

2.5 Neural Networks for Hyperbolic PDEs

Hyperbolic conservation laws pose specific challenges that have motivated a growing body of dedicated work. We organize this literature into three directions: hybrid numerical-neural methods, operator learning, and analysis of generalization.

Hybrid methods. One approach embeds neural networks directly into the finite volume framework. Lichtlé et al. [43] introduce Neural Finite Volume (NFV) methods that learn update rules over extended spatial-temporal stencils while preserving the conservation structure of the underlying scheme, achieving lower error than Godunov and ENO/WENO solvers. Lorin and Novruzı [46] propose a non-diffusive neural network method that decomposes the space-time domain into subdomains bounded by learned discontinuity lines, training separate networks for smooth regions and shock positions using PDE residual and Rankine–Hugoniot losses.

Operator learning. Wang and Townsend [72] construct the first rigorously justified probabilistic algorithm for recovering the solution operator of hyperbolic PDEs, combining randomized SVD with adaptive hierarchical domain partitioning to identify characteristics and approximate the Green’s function. Rodriguez-Torrado et al. [62] incorporate attention into a physics-informed

architecture for the Buckley–Leverett equation, where the attention mechanism helps adapt the network’s behavior to the nonlinear features of the solution near shock fronts. Huang et al. [23] combine operator learning with domain decomposition to generalize across different geometries, partitioning the physical domain into subdomains with a Schwarz-type inference procedure, though their analysis focuses on elliptic rather than hyperbolic problems.

Generalization analysis. Thodi et al. [66] provide an empirical study of generalization errors for physics-informed FNO solvers on the LWR traffic flow equation, finding that models can generalize beyond their training distribution, but with generalization error growing linearly with input complexity.

The present work differs from all of the above in its input representation and inductive bias: by operating on exact piecewise constant initial conditions rather than discretized grids, and by embedding characteristic speeds and Rankine–Hugoniot shock relations directly into the attention mechanism, WaveNO targets the specific structure of hyperbolic solutions rather than treating them as generic function approximation problems.

Chapter 3

Problem Formulation

This chapter establishes the mathematical framework that underpins the rest of the thesis. We begin with the general theory of hyperbolic conservation laws, the class of PDEs we aim to solve, and then specialize to the concrete equations used in our experiments: the LWR traffic flow model and its second-order extension ARZ. We close by formulating the operator learning problem: given a family of initial conditions, learn a map from initial data to the full space-time solution.

3.1 Scalar Conservation Laws

We consider the initial value problem (IVP) for a one-dimensional scalar conservation law,

$$\partial_t u + \partial_x f(u) = 0, \quad x \in \mathbb{R}, \quad t > 0, \quad (3.1)$$

with initial condition

$$u(0, x) = u_0(x), \quad (3.2)$$

where $u: [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$ is the conserved quantity and $f: \mathbb{R} \rightarrow \mathbb{R}$ is the flux function, assumed to be at least Lipschitz continuous. The flux determines the speed at which information propagates: the characteristic speed at a state u is

$$\lambda(u) = f'(u). \quad (3.3)$$

Along a curve $x(t)$ satisfying $\dot{x}(t) = f'(u(t, x(t)))$, the solution is constant in time, these curves are the characteristics of the equation, and they carry initial data forward into the space-time domain [35].

3.1.1 The Rankine–Hugoniot Condition

A fundamental property of (3.1) is that classical (C^1) solutions generally cease to exist in finite time: characteristics carrying different values of u may

intersect, forcing the solution to develop discontinuities (shocks) even from smooth initial data [35, 11]. At a discontinuity propagating along a curve $x = s(t)$, conservation across the shock implies the Rankine–Hugoniot jump condition [60, 24]:

$$\sigma[u] = [f(u)], \quad (3.4)$$

where $\sigma = \dot{s}(t)$ is the shock speed, and $[u] = u^+ - u^-$ denotes the jump across the discontinuity, with u^- and u^+ the left and right traces respectively. This condition determines the speed at which a shock must travel to conserve mass globally, and it plays a central role in both the numerical solution and the physics-informed losses used in this work.

3.1.2 The Riemann Problem

The Riemann problem is the IVP (3.1)–(3.2) with piecewise constant initial data consisting of a single discontinuity:

$$u_0(x) = \begin{cases} u_L & \text{if } x < 0, \\ u_R & \text{if } x > 0. \end{cases} \quad (3.5)$$

Its solution is self-similar, depending only on the ratio $\xi = x/t$ [35, 67]. For a scalar conservation law with convex or concave flux (f'' does not change sign), the Riemann solution consists of a single elementary wave:

- If $f'(u_L) > f'(u_R)$ (converging characteristics), a **shock** forms, propagating at speed $\sigma = \frac{f(u_R) - f(u_L)}{u_R - u_L}$.
- If $f'(u_L) < f'(u_R)$ (diverging characteristics), a **rarefaction wave** connects u_L to u_R through a smooth, self-similar fan.

The Riemann problem is the building block of the Godunov scheme [17] and, more importantly for our purposes, it determines the local wave structure at each discontinuity in a piecewise constant initial condition, the representation used throughout this thesis. Specifically, we restrict attention to initial conditions of the form

$$u_0(x) = \sum_{j=1}^K \rho_j \mathbf{1}_{[x_{j-1}, x_j)}(x), \quad (3.6)$$

where $x_0 < x_1 < \dots < x_K$ are the breakpoints and $\rho_1, \dots, \rho_K$ are the segment values. Each interior breakpoint x_j gives rise to a local Riemann problem between states ρ_j (left) and ρ_{j+1} (right).

3.1.3 The LWR Traffic Flow Model

The Lighthill–Whitham–Richards (LWR) model [44, 61] describes the evolution of traffic density $\rho(t, x) \in [0, \rho_{\max}]$ on a single-lane road as a scalar

conservation law (3.1) with flux

$$f(\rho) = \rho v(\rho), \quad (3.7)$$

where $v(\rho)$ is the velocity–density relationship (the “fundamental diagram”). The model assumes that drivers adjust their speed instantaneously to the local density, an equilibrium hypothesis that makes f a function of ρ alone.

The most widely used fundamental diagram is the Greenshields model [18], which posits a linear velocity–density relationship:

$$v(\rho) = v_{\max} \left(1 - \frac{\rho}{\rho_{\max}} \right), \quad \text{so that} \quad f(\rho) = v_{\max} \rho \left(1 - \frac{\rho}{\rho_{\max}} \right). \quad (3.8)$$

In normalized units ($v_{\max} = \rho_{\max} = 1$), this simplifies to

$$f(\rho) = \rho(1 - \rho), \quad f'(\rho) = 1 - 2\rho. \quad (3.9)$$

The flux is strictly concave ($f'' = -2 < 0$), so the entropy condition reduces to the Oleĭnik condition, and every discontinuity is either a shock or a rarefaction, no contact discontinuities arise. The characteristic speed $\lambda(\rho) = 1 - 2\rho$ ranges from $+1$ (free flow, $\rho = 0$) to -1 (jam, $\rho = 1$) and vanishes at the critical density $\rho_c = 1/2$, at which the flux achieves its maximum.

For a Riemann problem (3.5) with the Greenshields flux:

- **Shock** ($\rho_L < \rho_R$, i.e., $\lambda_L > \lambda_R$): propagates at speed $\sigma = 1 - \rho_L - \rho_R$.
- **Rarefaction** ($\rho_L > \rho_R$, i.e., $\lambda_L < \lambda_R$): a smooth fan connecting ρ_L to ρ_R through the self-similar variable $\xi = x/t$, with $\rho(\xi) = (1 - \xi)/2$.

The LWR model with Greenshields flux is the primary equation used in our experiments. Its simplicity makes it an ideal testbed: the concave flux produces rich shock interactions (merging, overtaking) from piecewise constant initial data, while the closed-form Riemann solution provides exact reference solutions for evaluation.

3.2 Systems of Conservation Laws

The scalar theory extends to systems. A one-dimensional system of m conservation laws reads

$$\partial_t \mathbf{u} + \partial_x \mathbf{f}(\mathbf{u}) = 0, \quad \mathbf{u}: [0, T] \times \mathbb{R} \rightarrow \mathbb{R}^m, \quad (3.10)$$

where $\mathbf{f}: \mathbb{R}^m \rightarrow \mathbb{R}^m$ is the flux vector. The system is hyperbolic if the Jacobian $\mathbf{f}'(\mathbf{u})$ has m real eigenvalues $\lambda_1(\mathbf{u}) \leq \dots \leq \lambda_m(\mathbf{u})$ with a complete set of eigenvectors for all physically relevant states [11, 36]. Each eigenvalue defines a *characteristic family*, and the solution of a Riemann problem consists of m

3.3. Operator Learning from Piecewise Constant Initial Conditions

waves, one per family, each of which is a shock, a rarefaction, or a contact discontinuity depending on the nonlinearity of that family.

The Rankine–Hugoniot condition generalizes to

$$\sigma[\mathbf{u}] = [\mathbf{f}(\mathbf{u})], \quad (3.11)$$

and entropy conditions must be imposed family by family. The theory is considerably richer and more delicate than in the scalar case, we refer to Dafermos [11] and Godlewski and Raviart [16] for a comprehensive treatment, but the essential structure is the same: information propagates along characteristics at finite speed, and discontinuities satisfy jump conditions that enforce conservation.

3.2.1 The ARZ Model

The Aw–Rascle–Zhang (ARZ) model [2, 78] is a 2×2 system of conservation laws that captures non-equilibrium traffic dynamics. Unlike LWR, drivers do not instantaneously adjust to the local density; instead, velocity evolves according to its own conservation law. The system reads

$$\partial_t \begin{pmatrix} \rho \\ \rho w \end{pmatrix} + \partial_x \begin{pmatrix} \rho v \\ \rho w v \end{pmatrix} = 0, \quad (3.12)$$

where ρ is density, v is velocity, and $w = v + p(\rho)$ is a generalized momentum that combines velocity with a “traffic pressure” $p(\rho)$. The pressure function $p(\rho) = \rho^\gamma$ (with $\gamma > 0$) models the anticipation of downstream conditions: drivers slow down when they perceive high density ahead.

The Jacobian of the flux has two distinct eigenvalues,

$$\lambda_1 = v - \rho p'(\rho), \quad \lambda_2 = v, \quad (3.13)$$

so the system is strictly hyperbolic whenever $\rho > 0$. The first family carries genuinely nonlinear waves (shocks and rarefactions, as in LWR), while the second family carries contact discontinuities across which ρ jumps but v is continuous. This richer wave structure makes ARZ a more challenging target for neural operators.

3.3 Operator Learning from Piecewise Constant Initial Conditions

The equations introduced above share a common structure: given an initial condition, they determine a unique solution on the space-time domain $[0, T] \times [0, L]$. We now formalize the task of learning this map from data.

3.3. Operator Learning from Piecewise Constant Initial Conditions

Let $\mathcal{U}_K$ denote the space of piecewise constant functions on $[0, L]$ with at most K segments, as defined in (3.6). Each $u_0 \in \mathcal{U}_K$ is fully characterized by its breakpoints $\mathbf{x} = (x_0, \dots, x_K)$ and segment values $\boldsymbol{\rho} = (\rho_1, \dots, \rho_K)$, a finite-dimensional representation that is exact regardless of the number of segments. Let $\mathcal{S}_T: \mathcal{U}_K \rightarrow L^\infty([0, T] \times [0, L])$ denote the solution operator that maps each initial condition to its solution on $[0, T]$.

Definition 3.1 (Operator learning problem) *Given a dataset $\{(u_0^{(i)}, u^{(i)})\}_{i=1}^N$ of piecewise constant initial conditions and corresponding reference solutions, find parameters θ of a neural operator $\hat{\mathcal{S}}_\theta$ that minimize*

$$\min_{\theta} \frac{1}{N} \sum_{i=1}^N \mathcal{L}(\hat{\mathcal{S}}_\theta(\mathbf{x}^{(i)}, \boldsymbol{\rho}^{(i)}), u^{(i)}), \quad (3.14)$$

where $\mathcal{L}$ is a loss functional on the space-time domain $[0, T] \times [0, L]$.

The reference solutions $u^{(i)}$ are evaluated on a uniform grid of $n_t \times n_x$ points. For scalar conservation laws with concave flux (LWR), they are computed exactly via the Lax–Hopf formula; for systems (ARZ), they are obtained from a high-resolution finite volume solver (HLL flux with WENO-5 reconstruction and SSP-RK3 time stepping). The neural operator maps the breakpoint representation directly to a prediction on the same grid:

$$\hat{\mathcal{S}}_\theta: (\mathbf{x}, \boldsymbol{\rho}) \longmapsto \hat{u} \in \mathbb{R}^{n_t \times n_x}. \quad (3.15)$$

This formulation departs from standard neural operator approaches. FNO [41] and DeepONet [48] discretize the initial condition on a fixed grid, discarding the exact locations of discontinuities. For hyperbolic conservation laws, this is particularly problematic: a discretization error of even half a cell width in a breakpoint position can shift a shock trajectory by $\mathcal{O}(T)$ in the output, since shocks propagate at finite speed over the entire time horizon. By operating on the exact breakpoints, our formulation eliminates this source of error entirely and provides a natural variable-length input: the same architecture handles initial conditions with different numbers of segments without retraining.

We evaluate generalization along two axes: (i) complexity, initial conditions with more breakpoints and richer wave interactions than those seen during training, and (ii) resolution, producing accurate solutions on finer output grids without retraining.

Architecture

This chapter describes the WaveNO architecture in detail. WaveNO realizes the neural operator map (3.15) through a five-stage pipeline: (1) encode each IC segment into a physics-augmented embedding, (2) contextualize embeddings via self-attention, (3) condition them on time with FiLM, (4) let space-time query points attend to segments via physics-biased cross-attention, and (5) project to density. The input is a piecewise constant initial condition of the form (3.6), represented by its breakpoints $\mathbf{x} = (x_0, \dots, x_K)$ and segment values $\boldsymbol{\rho} = (\rho_1, \dots, \rho_K)$. The output is a prediction $\hat{u} \in \mathbb{R}^{n_t \times n_x}$ on a space-time grid of query points (t_i, x_j) . Figure 4.1 provides a schematic overview of the full pipeline.

4.1 Segment Encoding

The first stage lifts each IC segment into a high-dimensional embedding that captures both its geometry and its role in the solution. A piecewise constant IC consists of K segments, each with constant density ρ_k on the interval $[x_k, x_{k+1}]$. A naive representation using only the raw breakpoints and values would miss derived physical quantities on which the solution critically depends: characteristic speeds determine how information propagates, flux values appear in the Rankine–Hugoniot condition (3.4), and the mass distribution constrains the solution globally.

For each segment k , the encoder computes three groups of features:

- **Geometric features** (equation-independent): center position $\bar{x}_k = (x_k + x_{k+1})/2$ and width $w_k = x_{k+1} - x_k$.
- **Physical features** (equation-specific): scalars derived from the state variables and the PDE’s flux function.

4.1. Segment Encoding

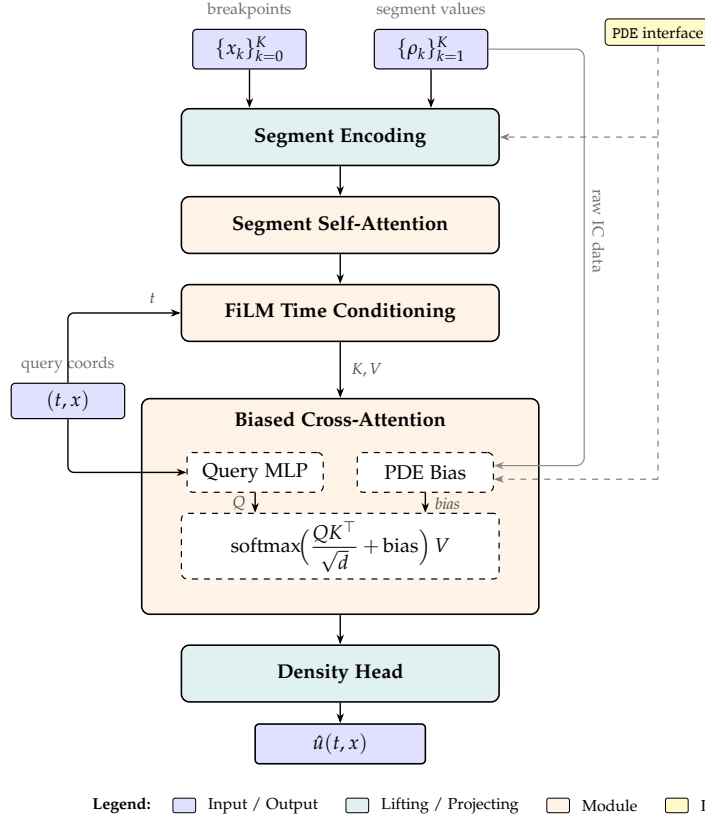


Figure 4.1: WaveNO architecture. Solid arrows show the main data flow; dashed arrows indicate physics quantities supplied by the pluggable PDE interface; the gray arrow carries raw IC data to the PDE bias module.

- **Global context:** normalized cumulative mass $N_k = \sum_{j < k} \rho_j w_j / M_{\text{total}}$, where $M_{\text{total}} = \sum_{j=1}^K \rho_j w_j$.

The physical features are provided by a pluggable PDE interface (described in section 4.7), so the encoder adapts automatically to the equation being solved. For the two equations used in our experiments:

- **LWR** (section 3.1.3): density ρ_k , characteristic speed $\lambda_k = f'(\rho_k)$, and flux value $f_k = f(\rho_k)$, giving 3 physical features per segment.
- **ARZ** (section 3.2.1): density ρ_k , velocity v_k , traffic pressure $p(\rho_k)$, and the first eigenvalue $\lambda_{1,k} = v_k - \rho_k p'(\rho_k)$, giving 4 physical features per segment.

The cumulative mass N_k encodes a conserved quantity of (3.1): total mass is an invariant, and providing each segment’s contribution helps the model respect conservation without an explicit constraint.

All features are concatenated and projected through an MLP with ReLU

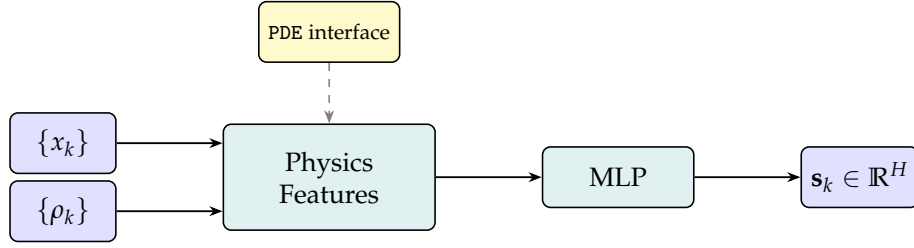


Figure 4.2: Segment encoding pipeline. Raw IC data (breakpoints and segment values) are augmented with physics features computed via the PDE interface, then projected through an MLP to produce segment embeddings $\mathbf{s}_k \in \mathbb{R}^H$.

activation and layer normalization to produce segment embeddings $\mathbf{s}_k \in \mathbb{R}^H$, where H is the hidden dimension. ReLU is used throughout the architecture: its piecewise-linear form is a natural match for the piecewise-constant initial conditions and the sharp discontinuities (shocks, contact waves) that characterize entropy solutions of (3.1), and deep ReLU networks have been shown to achieve optimal approximation rates for piecewise smooth functions, including discontinuous ones [54]. Figure 4.2 illustrates the encoding pipeline.

4.1.1 Segment Self-Attention

The solution at any point in the space-time domain depends on how all waves interact: a shock emanating from one discontinuity may collide with a rarefaction from another, producing structures that cannot be predicted from any single segment in isolation. For the model to anticipate these interactions, each segment must be aware of its neighbors' speeds, densities, and distances.

The segment embeddings pass through a stack of transformer encoder layers [71], each consisting of multi-head self-attention followed by a feed-forward network with ReLU activation, with residual connections and post-addition layer normalization [3]:

$$\mathbf{s} \leftarrow \text{LN}(\mathbf{s} + \text{MHA}(\mathbf{s}, \mathbf{s}, \mathbf{s})), \quad (4.1)$$

$$\mathbf{s} \leftarrow \text{LN}(\mathbf{s} + \text{FFN}(\mathbf{s})). \quad (4.2)$$

After self-attention, each segment embedding $\mathbf{s}_k$ encodes not only segment k 's local properties but contextual information about the entire IC, analogous to how a transformer encoder in language modeling lets each token attend to the full sequence.

4.2 Query Encoding

Each query point (t, x) on the output grid is encoded independently: the raw time and space coordinates are concatenated and projected through an MLP with ReLU activation to produce a query embedding $\mathbf{q}(t, x) \in \mathbb{R}^H$. This design follows the DeepONet paradigm [48]: a branch network encodes the input function (here the IC segments), while a trunk network encodes each evaluation point, and the two representations are combined to produce the output. In WaveNO, the combination is realized through cross-attention rather than a simple inner product, which allows each query to selectively attend to relevant segments. The query embeddings serve as the Q vectors in the subsequent cross-attention stage.

4.3 FiLM Time Conditioning

The IC defines the initial wave structure, but as time progresses, waves move, interact, and reshape the solution. A static segment embedding cannot capture this: the “meaning” of segment k at $t = 0$ (a constant region) differs from its meaning at a later time, when its wave has propagated and possibly collided with neighboring waves. The segment embeddings must therefore be evolved over time before they are used as keys and values in cross-attention. A natural approach would be autoregressive time-stepping, but this would force sequential computation across timesteps and prevent predicting the full space-time grid in a single forward pass. Instead, WaveNO uses Feature-wise Linear Modulation (FiLM) [53] to produce time-conditioned segment embeddings: each timestep modulates the IC embeddings independently, so all timesteps can be computed in parallel. An MLP takes the concatenation of the scalar time t and the segment embedding as input, and produces per-segment, per-timestep affine parameters:

$$\gamma(t, k), \beta(t, k) = \text{MLP}(t \parallel \mathbf{s}_k), \quad (4.3)$$

$$\mathbf{s}_k^{(t)} = \gamma(t, k) \odot \mathbf{s}_k + \beta(t, k), \quad (4.4)$$

where $\gamma, \beta \in \mathbb{R}^H$ and $\odot$ denotes element-wise multiplication. Because both time and the segment embedding are inputs to the FiLM network, the modulation is segment-specific: different segments can evolve at different rates (e.g., a segment behind a fast shock changes meaning faster than one in a quiescent region).

A key property of this design is that the modulation depends on time but not on space. The time-conditioned embeddings $\mathbf{s}_k^{(t)}$ represent the state of each segment at time t , and all spatial queries at the same timestep attend to the same set of conditioned segments. This guarantees spatial coherence: queries at a given time share a consistent view of the evolved IC, preventing

different spatial locations from receiving contradictory information about the segment states.

The FiLM parameters are initialized near identity ($\gamma = \mathbf{1}$, $\beta = \mathbf{0}$), so the model starts from the static baseline and gradually learns time dependence during training. The time-conditioned embeddings serve as keys and values in the per-timestep cross-attention that follows.

4.4 Physics-Informed Attention Bias

The physics-informed attention bias is the central inductive-bias component of WaveNO. It provides the cross-attention mechanism with a prior on which IC segments should influence each query point, computed from the analytical solution of the local Riemann problem at each interface in the initial condition. The construction is general: it applies to any scalar conservation law (3.1) with convex flux, requiring only the characteristic speed $f'(\cdot)$ and the Rankine–Hugoniot shock speed. We describe the mechanism in full generality and then instantiate it for the LWR model with Greenshields flux (3.9).

4.4.1 Motivation and Relation to ALiBi

In natural language processing, Press et al. [55] introduced ALiBi (Attention with Linear Biases), which adds a penalty proportional to the distance between query and key positions directly to the attention logits:

$$\text{bias}_{\text{ALiBi}}(i, j) = -m \cdot |i - j|, \quad (4.5)$$

where m is a head-specific slope and i, j are token positions. This acts as a soft positional prior: nearby tokens attend to each other more strongly than distant ones, without learned positional embeddings. Chi et al. [8] later showed that ALiBi’s linear penalty suppresses distant tokens so aggressively that it effectively limits the receptive field to a finite window. In natural language, where long-range dependencies are common, this is a significant limitation.

For hyperbolic PDEs, however, this locality property is not a limitation but a physical reality. The solution at a query point (t, x) depends only on the IC segments whose waves have physically reached that point by time t , a consequence of the finite propagation speed of (3.1). Masking distant segments is therefore not an approximation but the correct causal structure.

Concurrent work by Wu et al. [76] (FlashBias) demonstrates that adding a spatial distance bias to attention among mesh points improves PDE solving accuracy, confirming that distance-aware attention is beneficial in this domain. However, their bias is based on Euclidean distance between grid points, which treats all directions equally. In hyperbolic conservation laws, information

does not propagate isotropically: it travels along characteristics whose speeds depend on the local solution values. Our physics-informed bias goes further by computing the bias from the solution of the local Riemann problem at each interface, encoding not just spatial proximity but the actual wave speeds, shock trajectories, and rarefaction fans determined by the initial condition. Both ALiBi and the physics-informed bias are additive attention biases that softly modulate attention weights, but the latter replaces linear token distance (or Euclidean grid distance) with the wave propagation geometry of the PDE.

4.4.2 Wave Classification and Boundary Speeds

At each interior breakpoint $x_d = x_{k+1}$ between segments k and $k + 1$, a local Riemann problem (section 3.1.2) determines the wave structure. The bias classifies the wave type by comparing characteristic speeds (3.3):

$$\lambda_k > \lambda_{k+1} \implies \text{shock}, \quad \lambda_k \leq \lambda_{k+1} \implies \text{rarefaction}. \quad (4.6)$$

For a **shock**, the boundary between the two segments is a single trajectory moving at the Rankine–Hugoniot speed (3.4):

$$s = \frac{f(\rho_{k+1}) - f(\rho_k)}{\rho_{k+1} - \rho_k}. \quad (4.7)$$

Both segments are sharply cut off along this trajectory.

For a **rarefaction**, the fan occupies a wedge between the characteristic speeds λ_k and λ_{k+1} . Inside the fan, both segments have influence (the solution smoothly transitions between them), so each segment is penalized only beyond the far edge of the fan: segment k (on the left) is penalized past the right fan edge λ_{k+1} , and segment $k + 1$ (on the right) is penalized past the left fan edge λ_k .

The boundary speed for each side of the interface is therefore:

$$v_R = \begin{cases} s & \text{if shock,} \\ \lambda_{k+1} & \text{if rarefaction,} \end{cases} \quad v_L = \begin{cases} s & \text{if shock,} \\ \lambda_k & \text{if rarefaction,} \end{cases} \quad (4.8)$$

where v_R bounds the left segment’s influence zone to the right, and v_L bounds the right segment’s influence zone to the left.

4.4.3 Similarity Variable Formulation

The solution to a Riemann problem is self-similar: it depends only on the ratio $\xi = x/t$ (section 3.1.2). This motivates computing the bias in the self-similar coordinate rather than in physical space. For each interface at position x_d and each query point (t, x) , the similarity variable is

$$\xi = \frac{x - x_d}{t + \varepsilon}, \quad (4.9)$$

where $\varepsilon > 0$ is a small constant that regularizes the division at $t = 0$.

Working in ξ -space has three advantages over the physical-space alternative, where the penalty would take the form $\text{ReLU}((x - x_d) - v_R t)$.

Time-invariant influence boundaries. In physical space, the boundary of a segment's influence zone is the line $x = x_d + v_R t$, which moves through the domain as t grows. In ξ -space, the same boundary is the fixed threshold $\xi = v_R$, independent of t . The penalty function therefore has the same shape at every time step, giving the cross-attention a consistent, time-invariant structure that does not need to be learned.

Scale invariance. Scalar conservation laws are invariant under the hyperbolic scaling $(x, t) \rightarrow (\alpha x, \alpha t)$. The physical-space penalty scales with α , so a model trained on one domain size would see different bias magnitudes at test time on a larger or smaller domain. Because $\xi = x/t$ is itself invariant under this scaling, the ξ -space penalty is unchanged, and the bias generalizes across domain sizes and time horizons without rescaling.

Implicit time-dependent prior strength. The two formulations are related by $\text{ReLU}(\xi - v_R) = \text{ReLU}((x - x_d) - v_R t) / (t + \varepsilon)$: the ξ -space penalty is the physical-space penalty divided by t . Since the bias is added to the attention logits before the softmax, this $1/t$ scaling creates an implicit time-dependent prior strength: at early times (small t) the penalty magnitudes are large and the physics prior dominates the attention weights, while at later times (large t) the magnitudes shrink and the learned attention can override the prior. This is desirable because the Riemann-based bias is exact before wave interactions and becomes unreliable after neighboring waves collide, so the prior should be strongest precisely when it is most accurate.

The one-sided penalties for each interface are:

$$\text{penalty}_L(\xi, k) = \text{ReLU}(\xi - v_R), \quad (4.10)$$

$$\text{penalty}_R(\xi, k+1) = \text{ReLU}(v_L - \xi), \quad (4.11)$$

where penalty_L penalizes the left segment k when the query lies to the right of its influence boundary, and penalty_R penalizes the right segment $k+1$ when the query lies to the left. The ReLU ensures the penalty is zero inside the influence zone and grows linearly with the excess similarity variable outside it.

Each segment accumulates penalties from both its left and right interfaces. Denoting the similarity variable at interface d by $\xi_d = (x - x_d) / (t + \varepsilon)$, the total bias for segment k is

$$\text{bias}(t, x, k) = -\text{penalty}_L(\xi_k, k) - \text{penalty}_R(\xi_{k-1}, k), \quad (4.12)$$

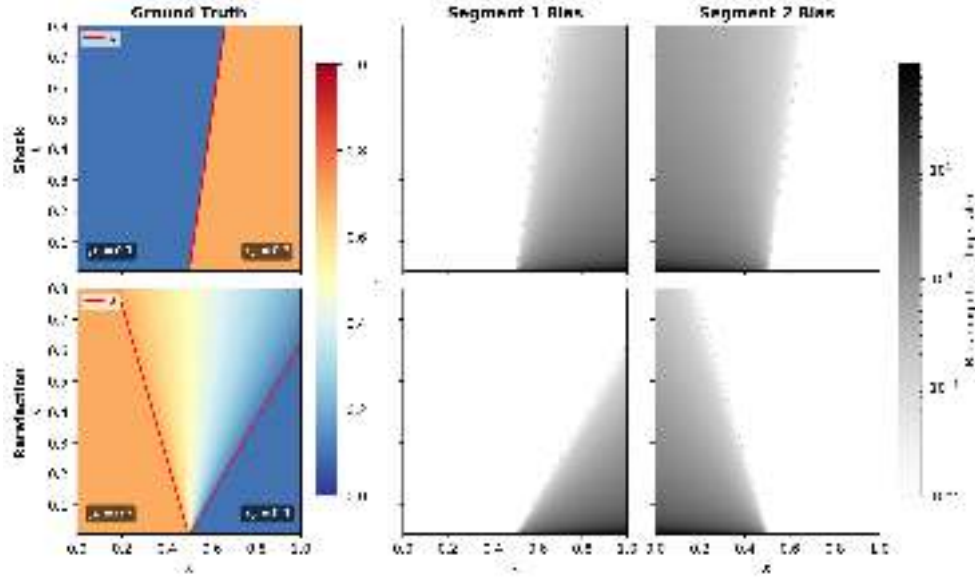


Figure 4.3: Physics-informed attention bias for the LWR Greenshields flux, shown on a logarithmic color scale for a two-segment IC with interface at $x_d = 0.5$. **Top row (shock):** $\rho_L = 0.1, \rho_R = 0.7$; the shock propagates at speed $s = 0.2$. **Bottom row (rarefaction):** $\rho_L = 0.7, \rho_R = 0.1$; the fan opens between $\lambda_L = -0.4$ and $\lambda_R = 0.8$. Both segments retain influence inside the fan. The $1/t$ scaling is visible: the bias reaches ~ -50 near $t = 0$ and decays at later times.

where the first term uses interface k (right boundary of segment k) and the second uses interface $k-1$ (left boundary). The result is zero where the segment has physical influence and increasingly negative elsewhere.

For the LWR model with Greenshields flux (3.9), the characteristic speed is $\lambda(\rho) = 1 - 2\rho$ and the shock speed is $s = 1 - \rho_L - \rho_R$, so all boundary speeds in (4.8) are available in closed form. Figure 4.3 visualizes the resulting bias for a two-segment Riemann problem on a logarithmic color scale. The bias is zero (white) inside each segment's influence zone and increasingly negative (dark) beyond the wave trajectory. In the rarefaction case, both segments retain influence inside the fan, with penalties starting only beyond the far fan edge. The $1/t$ magnitude scaling is visible: the bias reaches values of order -50 near $t = 0$ and decays at later times, illustrating the implicit time-dependent prior strength described above.

4.4.4 Extension to the ARZ System

The ARZ system (section 3.2.1) has two characteristic families at each interface: a genuinely nonlinear 2-wave (which can be a shock or rarefaction) and a linearly degenerate 1-contact discontinuity propagating at speed v_R . Depending on the left and right states, three cases arise:

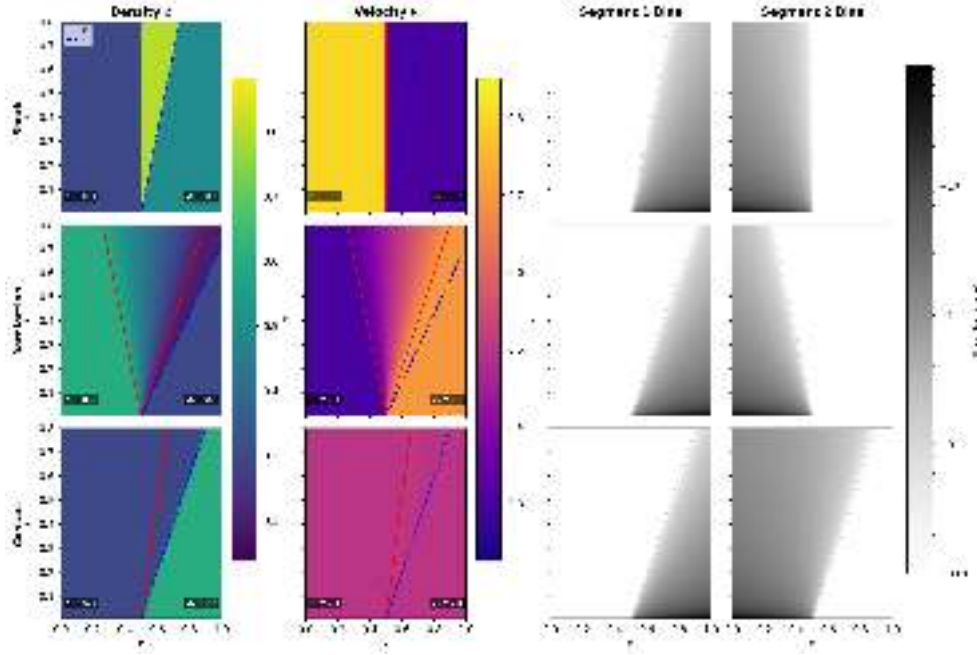


Figure 4.4: Physics-informed attention bias for the ARZ system with $\gamma = 1$, shown on a logarithmic color scale for a two-segment IC with interface at $x_d = 0.5$. **Top row (shock):** $\rho_L = 0.3$, $v_L = 0.8$, $\rho_R = 0.5$, $v_R = 0.3$; the 2-shock (red) and 1-contact (blue dashed) bound the segments. **Middle row (rarefaction):** $\rho_L = 0.6$, $v_L = 0.3$, $\rho_R = 0.3$, $v_R = 0.7$; both boundaries follow the 2-wave fan edges. **Bottom row (contact):** $\rho_L = 0.3$, $v_L = 0.5$, $\rho_R = 0.6$, $v_R = 0.5$; only the 1-contact propagates, bounding both segments.

- **Shock** ($v_L > v_R$): the 2-wave is a shock with Rankine–Hugoniot speed σ_2 . The left segment is bounded by the 1-contact at speed v_R and the right segment by the 2-shock at speed σ_2 .
- **Rarefaction** ($v_L < v_R$): the 2-wave is a rarefaction fan between the eigenvalues $\lambda_2(\rho_L, v_L)$ and $\lambda_2(\rho^*, v_R)$. Both segment boundaries are determined by the fan edges, ignoring the 1-contact.
- **Contact discontinuity** ($v_L = v_R$): the 2-wave is trivial ($\rho^* = \rho_L$) and only the 1-contact propagates. Both segment boundaries use the contact speed v_R .

Figure 4.4 visualizes the ARZ bias for all three wave types. In the shock case, the bias cones are bounded by the 2-shock (red) on the left and the 1-contact (blue) on the right. In the rarefaction case, both boundaries follow the 2-wave fan edges. In the contact case, both boundaries coincide with the 1-contact trajectory.

4.5 Biased Cross-Attention

At this stage, each query point (t, x) must determine what density value it should take. The answer depends on which IC segments influence that point and how their contributions combine. The cross-attention mechanism handles this by letting each query attend to all time-conditioned segment embeddings, steered by the physics-informed bias.

The query embeddings serve as queries, and the time-conditioned segment embeddings $\mathbf{s}_k^{(t)}$ from (4.4) serve as keys and values. The attention computation follows the standard scaled dot-product formulation [71] with an additive bias term:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}} + \text{bias}\right) V, \quad (4.13)$$

where $d = H/n_{\text{heads}}$ is the per-head dimension. The bias acts as a soft prior: it makes it easy for the model to attend to the physically relevant segments (those with bias = 0) and hard to attend to irrelevant ones (those with large negative bias), but the learned $QK^\top$ scores can override the prior when needed, for instance after wave collisions when the initial Riemann structure has been disrupted.

Each cross-attention layer uses a pre-normalization architecture [3], in contrast to the post-normalization used in self-attention, to stabilize training when the bias term introduces large-magnitude logits:

$$\mathbf{q}' \leftarrow \mathbf{q} + \text{MHA}(\text{LN}(\mathbf{q}), \text{LN}(\mathbf{s}^{(t)}), \text{LN}(\mathbf{s}^{(t)}), \text{bias}), \quad (4.14)$$

$$\mathbf{q} \leftarrow \mathbf{q}' + \text{FFN}(\text{LN}(\mathbf{q}')), \quad (4.15)$$

where the feed-forward network uses ReLU activation. Multiple biased cross-attention layers are stacked, allowing iterative refinement of query representations. The cross-attention operates per-timestep: at each time t_i , query embeddings attend to the segment embeddings for that timestep.

4.6 Density Head

After cross-attention, each query embedding encodes which segments are relevant and what their time-evolved properties are. The output head converts this contextual representation to a prediction through an MLP:

$$\hat{u}(t, x) = \text{MLP}(\mathbf{q}(t, x)) \in \mathbb{R}^d, \quad (4.16)$$

where d is the number of output channels, determined by the PDE: $d = 1$ for scalar equations (LWR) and $d = 2$ for systems (ARZ, predicting density

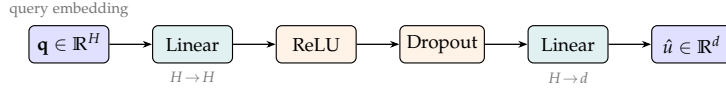


Figure 4.5: Output head pipeline. The query embedding is passed through an MLP to produce the predicted state $\hat{u} \in \mathbb{R}^d$.

and velocity). The output dimension, along with any equation-specific post-processing (e.g., clamping to physically admissible ranges), is specified by the PDE interface described in section 4.7. Figure 4.5 illustrates the output head pipeline.

4.7 Adaptability to Other PDEs

A central design goal of WaveNO is that the core architecture, segment encoding, self-attention, FiLM conditioning, biased cross-attention, and the output head, remains unchanged across equations. All equation-specific knowledge is isolated in a single abstract PDE interface that each conservation law implements. The interface exposes three methods:

1. **Physics features.** Given the IC segment data (state variables, breakpoints), return a tensor of per-segment scalar features for the segment encoder (section 4.1). For LWR this produces $[\rho_k, f'(\rho_k), f(\rho_k)]$; for ARZ it produces $[\rho_k, v_k, p(\rho_k), \lambda_{2,k}]$.
2. **Boundary speeds.** Given the IC segment data, return the right-boundary and left-boundary speeds at each interface for the physics-informed attention bias (section 4.4). For a scalar conservation law, these are the shock speed or the rarefaction fan edges (4.8). For a system, each characteristic family contributes its own boundary speed.
3. **Output metadata.** The number of output channels d and, optionally, a clamp range for physically admissible values (e.g., $\rho \in [0, 1]$ for LWR).

Adding a new equation requires only implementing these three methods in a new subclass. No changes to the model, the training loop, or the loss are needed: the segment encoder reads the number of physics features from the interface and adjusts its input dimension automatically, the bias module calls the boundary-speed method, and the output head reads the output dimension. This plug-and-play design ensures that the architecture can be extended to new conservation laws with minimal effort, while keeping all equation-specific analytical knowledge, characteristic speeds, shock speeds, and state-variable definitions, outside the learned components.

Experiments

Having described the WaveNO architecture in chapter 4, we now evaluate it empirically. The experiments pursue two goals: first, to compare WaveNO against established neural operator baselines on two hyperbolic conservation laws of increasing complexity, and second, to ablate the main architectural components and quantify their individual contributions. We focus on the LWR traffic flow model (section 3.1.3) and the ARZ system (section 3.2.1), which together cover scalar and system conservation laws with shocks, rarefactions, and contact discontinuities.

5.1 Experimental Setup

5.1.1 Equations and Data Generation

We evaluate on two equations. The **LWR model** with Greenshields flux (3.9) is a scalar conservation law whose concave flux produces rich shock interactions from piecewise constant initial data. Reference solutions are computed with the Lax–Hopf formula, which yields exact entropy solutions for scalar conservation laws with convex or concave flux, so evaluation is against ground truth rather than a numerical approximation. The **ARZ model** (3.12) is a 2×2 system with genuinely nonlinear and linearly degenerate families, producing shocks, rarefactions, and contact discontinuities. Reference solutions are computed with a high-resolution finite volume solver (HLL flux, WENO-5 reconstruction, SSP-RK3 time stepping).

For both equations, initial conditions are piecewise constant functions of the form (3.6), generated by sampling random breakpoint positions and segment values within the physically admissible range. The number of segments varies by equation: 2 to 8 for LWR and 2 to 6 for ARZ during training, with out-of-distribution testing extending to 16 and 12 segments, respectively. The computational domain is $[0, L] \times [0, T]$ with $L = 1$, $T = 1$ s, discretized on a

uniform grid of $n_x = 50$ spatial points and $n_t = 250$ time steps. We generate 4000 training samples per equation.

5.1.2 Models and Ablations

We compare five learned models organized into two groups.

Baselines. **FNO** [41] operates in the Fourier domain, learning global convolution filters via spectral truncation. **DeepONet** [48] uses a branch-trunk architecture, encoding the IC with a branch network and evaluation points with a trunk network, combining them via inner product. Both baselines receive the initial condition discretized on the $n_x = 50$ spatial grid, discarding the exact breakpoint locations, but inherently able to interpolate them since we don't allow discontinuities in the initial conditions to be within dx of one another.

WaveNO variants. The three WaveNO variants form a progressive ablation of the full architecture described in chapter 4:

- **WaveNO-Bare:** cross-attention between query points and segment embeddings, without the physics-informed bias or FiLM conditioning. Segment embeddings are static across time steps.
- **WaveNO-Bias:** adds the physics-informed attention bias (section 4.4) to the cross-attention logits (4.13), providing a wave-geometry prior on which segments should influence each query point.
- **WaveNO-Full:** further adds FiLM time conditioning (section 4.3), which evolves segment embeddings over time via learned affine modulation (4.4).

All three variants receive the exact breakpoints and segment values as input, bypassing discretization entirely. For LWR, we also include the first-order **Godunov scheme** [17] evaluated on the same 50×250 grid as a numerical reference point.

5.1.3 Training and Evaluation Protocol

All models are trained for 100 epochs with the MSE loss (3.14) using the Adam optimizer. Each model-equation pair is trained with three random seeds. The primary evaluation metric is the mean squared error (MSE) on the held-out test set. Full training details are given in the appendix.

Table 5.1: Test MSE (mean $\pm$ std over 3 seeds). Best learned model per equation in bold. Godunov is a numerical solver, not a learned model.

Model	LWR	ARZ
FNO	0.0027 ± 0.0004	0.0020 ± 0.0003
DeepONet	0.0046 ± 0.0005	0.0041 ± 0.0002
Godunov	0.0020 ± 0.0001	—
WaveNO-Bare	0.0011 ± 0.0001	0.0016 ± 0.0001
WaveNO-Bias	0.0011 ± 0.0001	0.0016 ± 0.0001
WaveNO-Full	0.0010 ± 0.0001	0.0014 ± 0.0001

5.2 Overall Accuracy

Table 5.1 reports the test MSE for all models on both equations. WaveNO-Full achieves the lowest MSE on both LWR and ARZ. On LWR, it reduces the MSE by 63.4% relative to FNO and by 78.8% relative to DeepONet. On ARZ, the improvements are 30.2% and 66.1%, respectively. Even WaveNO-Bare, the simplest variant without bias or FiLM, already outperforms both baselines on every equation, demonstrating that the cross-attention mechanism operating on exact breakpoints is itself a substantial advantage over grid-based input representations.

On LWR, WaveNO-Full (MSE 0.0010) also outperforms the Godunov scheme (MSE 0.0020). This comparison requires careful interpretation: Godunov is a first-order finite volume method operating on the same 50×250 grid, so it suffers from numerical diffusion that smears sharp discontinuities. The ground truth is the exact Lax–Hopf solution, not the Godunov approximation. The neural operator can outperform Godunov precisely because it is not constrained to a local stencil and can learn global solution structure from data.

Figure 5.2 shows the full pairwise improvement matrix for both equations. Each cell reports the percentage MSE reduction of the row model relative to the column model. On LWR, the WaveNO variants improve over FNO by 58–63% and over DeepONet by 76–79%. On ARZ, the pattern is consistent but the margins are smaller (17–30% over FNO, 60–66% over DeepONet), reflecting the greater difficulty of the system case.

5.3 Ablation: Component Contributions

The three WaveNO variants isolate the contribution of each architectural component. WaveNO-Bare uses only cross-attention on exact breakpoints, WaveNO-Bias adds the physics-informed bias, and WaveNO-Full further

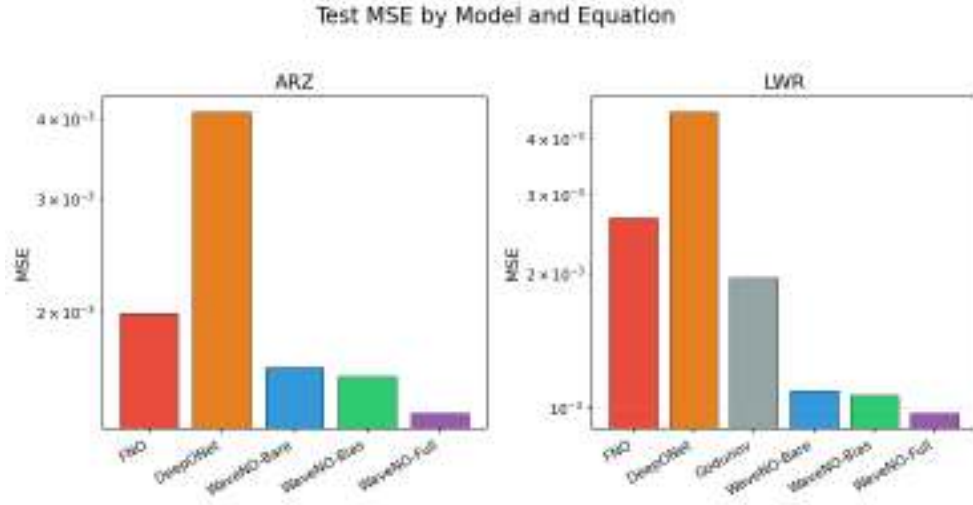


Figure 5.1: Test MSE by model and equation (log scale, mean over 3 seeds). WaveNO variants consistently outperform both baselines, with WaveNO-Full achieving the lowest error on LWR and ARZ.

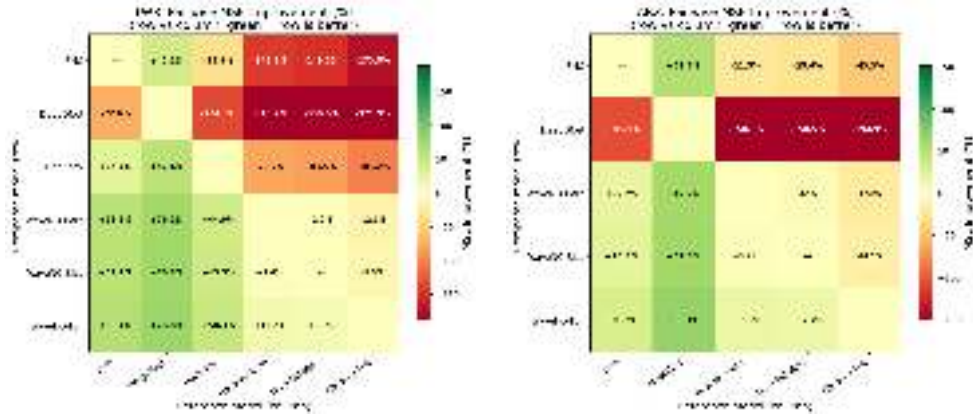


Figure 5.2: Pairwise MSE improvement (%) for LWR (left) and ARZ (right). Each cell shows the percentage MSE reduction of the row model over the column model. Green indicates the row model is better.

adds FiLM time conditioning. This section analyzes the incremental benefit of each addition.

5.3.1 Architecture Progression

Figure 5.3 shows the MSE at each stage of the progression. On LWR, the bias reduces the error by 2.6% relative to Bare, and FiLM conditioning adds a further 8.8%, for a cumulative improvement of 11.2% from Bare to Full. On ARZ, the progression is similar: 3.2% from the bias and 12.4% from FiLM,

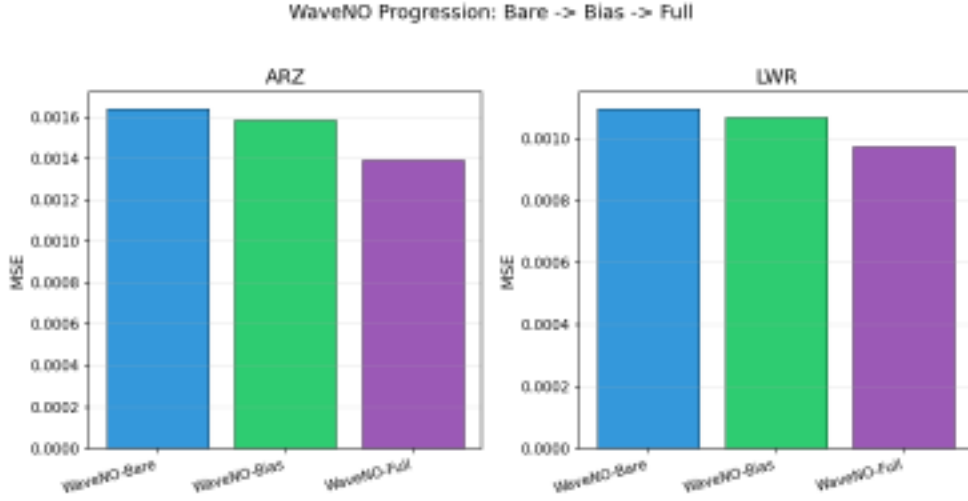


Figure 5.3: Incremental MSE reduction from Bare to Bias to Full. Percentages indicate the relative improvement at each step. The progression is consistent on LWR and ARZ.

totaling 15.2%. (All percentages are computed from unrounded seed-level values; differences between Bare and Bias that appear identical in table 5.1 become visible at higher precision.)

The bias contribution is smaller in magnitude than the gains from the cross-attention itself (which already reduces error by 58% over FNO on LWR). This is expected: the cross-attention is the primary mechanism for relating queries to segments, while the bias provides a refinement that steers attention toward physically relevant segments. FiLM conditioning contributes more than the bias on both equations, suggesting that evolving segment representations over time is important for capturing post-interaction dynamics that the static bias cannot encode.

5.3.2 Temporal Error Analysis

A key prediction of the architecture is that the physics-informed bias should provide its strongest advantage at early time steps. The bias is computed from local Riemann problems (section 4.4.2) and is exact before waves from neighboring interfaces interact. Moreover, the similarity-variable formulation (section 4.4.3) introduces an implicit $1/t$ scaling: at small t , the bias magnitudes are large and the physics prior dominates the attention weights, while at later times the magnitudes decay and the learned scores can override the prior.

Figure 5.4 plots the per-timestep MSE across the full time horizon for LWR and ARZ. Two patterns emerge. First, FNO starts with relatively low error at

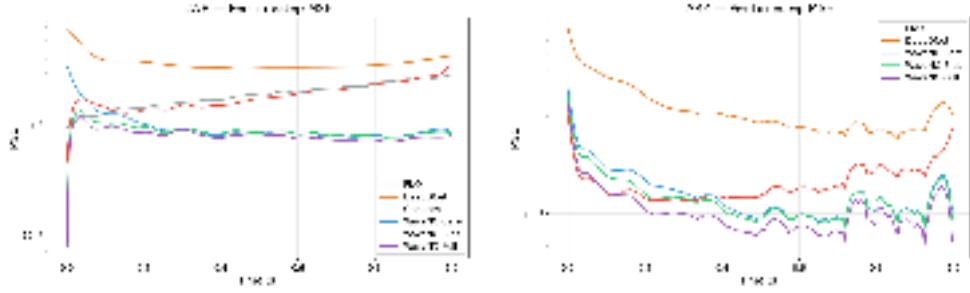


Figure 5.4: Per-timestep MSE (mean over 3 seeds) for LWR (left) and ARZ (right). Encoder-decoder models (DeepONet, WaveNO variants) maintain flat error profiles, while FNO degrades significantly over the time horizon.

early times but degrades steadily across the time horizon, reaching an $8\times$ increase on LWR from its minimum to $t = T$. In contrast, both DeepONet and all WaveNO variants maintain nearly flat MSE throughout. This reflects a fundamental difference between encoder-decoder and spectral architectures: encoder-decoder models evaluate each query point by combining it with an encoded representation of the initial condition, so each space-time location is predicted independently and errors at late times do not compound from earlier ones. FNO, on the other hand, applies global Fourier convolutions that process the entire grid jointly, causing spectral artifacts to accumulate across the time dimension and smear sharp features progressively.

Second, fig. 5.5 isolates the effect of the physics-informed bias by plotting the per-timestep percentage MSE improvement of WaveNO-Bias over WaveNO-Bare (excluding $t = 0$, where the bias trivially dominates due to the $1/t$ scaling). On LWR, the bias provides a clear advantage at early times (20–30% improvement for $t < 0.1$ s), which gradually decreases as the time horizon progresses and wave interactions invalidate the local Riemann structure. The improvement stabilizes around 5% at later times, indicating that the bias remains mildly helpful even after interactions, likely because some segments still have intact influence zones. On ARZ, the pattern is similar but noisier, with the bias providing 10–20% improvement in the early phase. This temporal profile matches the prediction: the bias is most valuable precisely when the local Riemann structure it encodes is still accurate, and its contribution fades as the solution departs from the initial wave geometry.

5.4 Generalization

A central claim of the WaveNO design is that operating on exact breakpoints and using only cross-attention and no self-attention among queries yields resolution-independent predictions (section 6.1.3). This is reinforced by the use of linear query encodings rather than Fourier features (section 6.1.2):

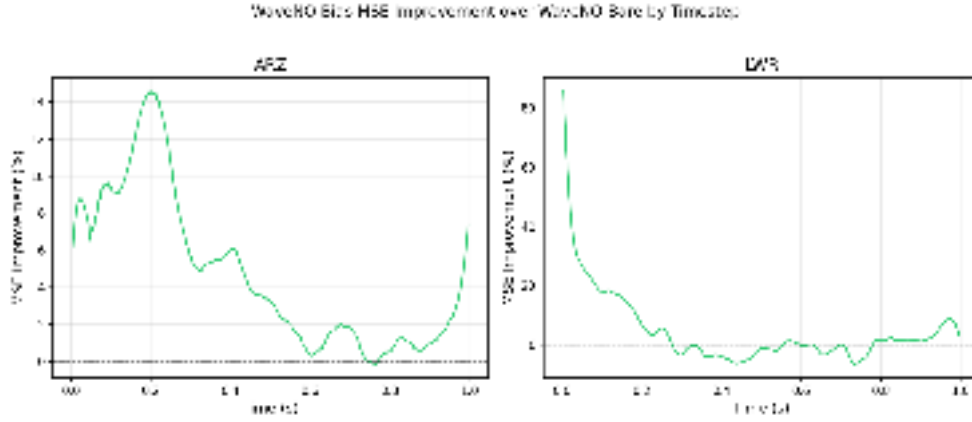


Figure 5.5: Per-timestep MSE improvement (%) of WaveNO-Bias over WaveNO-Bare, excluding $t = 0$. The bias provides its largest advantage at early times, when the local Riemann structure is still intact, and its contribution fades as wave interactions progress.

neural networks with smooth activations naturally interpolate smoothly on unseen inputs, so querying at new grid points produces well-behaved predictions. Fourier features would counteract this by introducing fixed-frequency sinusoidal bases whose oscillation peaks shift with the grid spacing, creating spurious high-frequency artifacts at unseen resolutions. We test these claims by evaluating models trained at the base resolution (50×250) on finer grids, and by varying the complexity of the initial condition at test time.

5.4.1 Resolution Generalization

Figure 5.6 shows the test MSE for all learned models as a function of the output resolution multiplier, from $1 \times$ (training resolution) to $5 \times$. All models are trained at the base resolution (50×250) and evaluated at finer grids without retraining. The two equations exhibit qualitatively different behavior, reflecting a fundamental difference in how their reference solutions are generated.

LWR (exact ground truth). Because LWR reference solutions are computed with the Lax–Hopf formula, the ground truth is exact at every resolution: increasing the grid density does not change the target function, only the number of points at which it is sampled. The WaveNO variants remain nearly flat across all resolutions, confirming that the cross-attention architecture generalizes smoothly to unseen grid spacings. FNO’s error actually *decreases* at higher resolutions (from 0.0027 at $1 \times$ to 0.0014 at $3 \times$), likely because its fixed Fourier modes represent lower physical frequencies on a finer

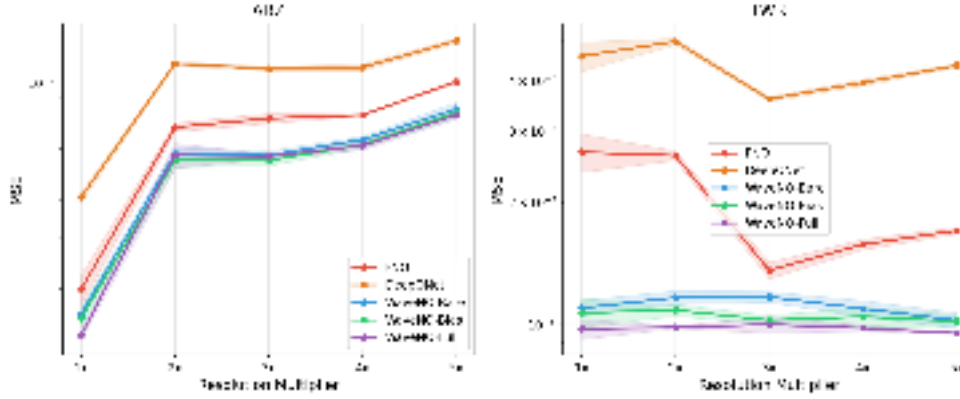


Figure 5.6: Test MSE as a function of the output resolution multiplier (mean $\pm$ std over 3 seeds, log scale). All models are trained at $1\times$ resolution. On LWR (exact ground truth), WaveNO variants remain flat and FNO improves. On ARZ (numerical ground truth), all models degrade as the reference solutions become sharper at finer grids.

grid, suppressing the oscillatory artifacts that dominate the error at training resolution. DeepONet fluctuates without a clear trend.

ARZ (numerical ground truth). ARZ reference solutions are obtained from a finite volume solver whose accuracy improves with the grid: finer meshes produce sharper shocks and thinner contact discontinuities, yielding a more demanding target. Consequently, *all* models degrade at higher resolutions, including the WaveNO variants (which increase from 0.0014 at $1\times$ to 0.0078 at $5\times$) and FNO (from 0.0020 to 0.0102). This degradation reflects a confound in the evaluation rather than a failure of resolution invariance: the models are compared against a progressively better reference solution, not merely queried at new grid points. The LWR results, where the ground truth is resolution-independent, provide the cleaner test of resolution generalization.

5.4.2 Initial Condition Complexity

The training data contains ICs with 2 to 8 segments for the LWR, 2 to 6 for the ARZ. Figure 5.7 stratifies the WaveNO-Bias and WaveNO-Full improvement over WaveNO-Bare by the number of IC segments, serving as a proxy for out-of-distribution complexity.

On LWR and ARZ, the improvement of the biased variants over Bare is largest for simple ICs (few segments) and decreases as the segment count grows. For ICs with many segments, the bias variants offer little advantage and can occasionally underperform Bare, particularly on ARZ.

This behavior has a clear interpretation. The physics-informed bias is com-

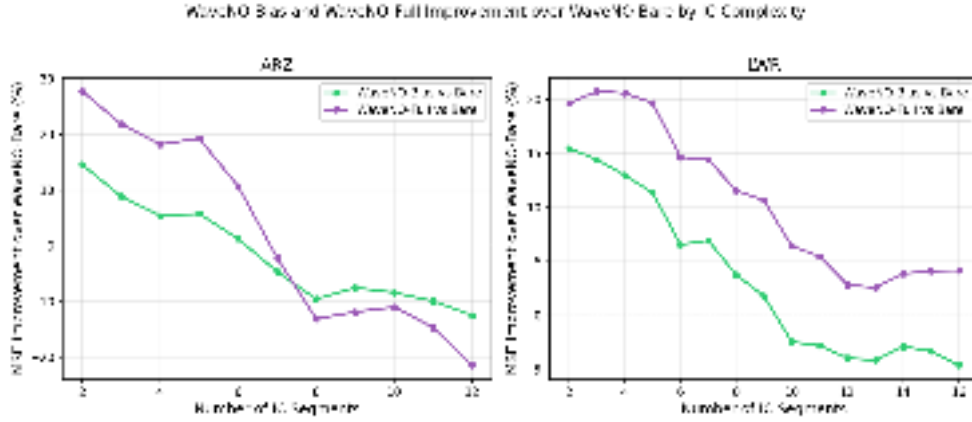


Figure 5.7: MSE improvement (%) of WaveNO-Bias and WaveNO-Full over WaveNO-Bare as a function of the number of IC segments. The advantage of the biased variants decreases with IC complexity, as wave interactions invalidate the local Riemann prior earlier. For OOD segment counts, Bare can be preferable.

puted from local Riemann problems at each interface and is exact before neighboring waves interact. With few segments, the waves remain isolated for a longer portion of the time horizon, and the bias provides an accurate prior throughout. With many segments, waves interact early and the local Riemann structure breaks down quickly, making the bias inaccurate for most of the solution. In this regime, the cross-attention must learn to override the bias via the $QK^\top$ scores (4.13). For segment counts not seen during training (out-of-distribution), the model has not learned this override pattern, so the bias can become actively harmful. WaveNO-Bare, which has no bias to override, does not suffer from this failure mode. This trade-off motivates the shared-bias design discussed in section 6.1.1: per-head slopes would further specialize the bias to the training distribution, exacerbating the OOD degradation.

5.5 Qualitative Analysis

Figure 5.8 shows spatial density profiles for a representative LWR test sample at five time fractions. At $t = 0$, the ground truth is a sharp step function. WaveNO-Bias and WaveNO-Full reproduce the step almost exactly, consistent with their near-zero MSE at $t = 0$ reported in fig. 5.4. WaveNO-Bare and FNO produce visible oscillations near the discontinuity, while DeepONet shows a broad smearing of the step.

As time progresses, the solution develops shocks and rarefactions. At $t = T/4$ and $t = T/2$, the WaveNO variants capture the shock fronts with sharp transitions, while FNO and DeepONet exhibit Gibbs-like oscillations

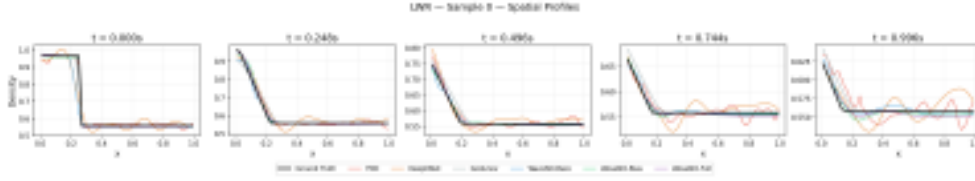


Figure 5.8: Spatial density profiles for LWR test sample 0 at five time fractions ($t = 0, T/4, T/2, 3T/4, T$). Ground truth in black. WaveNO-Bias and WaveNO-Full capture sharp discontinuities, especially at early times, while FNO and DeepONet exhibit oscillations and smearing.

near discontinuities. By $t = T$, all models track the general shape of the solution, but FNO displays large-amplitude oscillations in smooth regions. The WaveNO variants maintain cleaner profiles throughout, with the full model producing the sharpest shock edges.

Additional spatial profiles for ARZ and other test samples are provided in the appendix.

5.6 Computational Efficiency

Table 5.2 reports the parameter count and mean inference time per sample for each model. WaveNO-Full has 226K parameters, smaller than FNO (599K) and comparable to WaveNO-Bare (214K), confirming that the bias and FiLM modules add minimal overhead in model size.

The inference time of WaveNO-Full (17.1 ms) is higher than FNO (2.4 ms) and DeepONet (3.0 ms). The bottleneck is the cross-attention, which requires $O(n_t \cdot n_x \cdot K)$ operations per layer, where K is the number of IC segments. However, WaveNO evaluates the full space-time grid in a single forward pass, whereas autoregressive models require sequential evaluation over time steps. Moreover, FNO and DeepONet are an order of magnitude faster than the Godunov solver (31.8 ms on the same grid), and WaveNO-Full is roughly twice as fast, despite the Godunov scheme iterating sequentially over all n_t time steps.

Table 5.2: Model size and inference time. WaveNO-Full achieves the best accuracy with fewer parameters than FNO, at the cost of higher inference time due to cross-attention.

Model	Parameters	Inference time (ms)
FNO	599K	2.4 ± 0.0
DeepONet	135K	3.0 ± 0.0
Godunov	—	31.8 ± 0.1
WaveNO-Bare	214K	13.1 ± 0.2
WaveNO-Bias	214K	16.8 ± 5.4
WaveNO-Full	226K	17.1 ± 5.6

Discussion

6.1 Architectural Design Choices

This section discusses four design decisions in the WaveNO architecture, each contrasting the chosen approach with a natural alternative and explaining why the alternative was rejected.

6.1.1 Shared Bias vs. Per-Head Slopes

ALiBi [55] assigns each attention head a different slope m , encouraging different heads to specialize in different ranges of token distance. A direct adaptation to WaveNO would multiply the physics-informed bias by a learnable per-head scalar m_h , giving each head a different “sharpness” of the influence zone boundary:

$$\text{bias}_h(t, x, k) = m_h \cdot \text{bias}(t, x, k). \quad (6.1)$$

We instead broadcast the same bias to all heads, i.e. $m_h = 1$ for every h .

The reason is out-of-distribution robustness. With per-head slopes, the model partitions the K segments across heads during training: a head with large m_h attends almost exclusively to the nearest segment, while a head with small m_h distributes attention more broadly. This partition is learned from the training distribution of segment counts and widths. At test time, when the number of segments or their spacing differs from training, the learned partition may no longer be appropriate, and heads that were trained to attend narrowly may miss relevant segments entirely. With a shared bias, all heads see the same physics-informed prior, and the learned $QK^\top$ scores are solely responsible for differentiating head behavior. Since the bias magnitude depends only on the wave geometry (characteristic speeds and shock speeds), it generalizes to unseen ICs without relearning head-specific slopes.

6.1.2 Linear Query Encoding vs. Fourier Features

Fourier feature encodings [65] map low-dimensional coordinates to high-dimensional sinusoidal representations, helping MLPs learn high-frequency functions. A natural choice for query encoding would be to apply Fourier features to the (t, x) coordinates before the query MLP.

However, Fourier features introduce resolution dependence. The sinusoidal basis functions $\sin(2\pi \cdot 2^f \cdot x)$ and $\cos(2\pi \cdot 2^f \cdot x)$ oscillate at fixed spatial frequencies. When the model is trained on a grid of resolution n_x and tested on a finer grid $n'_x > n_x$, the higher-frequency basis functions sample new oscillation peaks that were not present in the training grid. This creates spurious high-frequency patterns in the query embeddings, which propagate through cross-attention and produce oscillatory artifacts in the predicted density field.

The default WaveNO passes raw (t, x) coordinates through a linear projection followed by an MLP. Because linear projections are smooth functions of their inputs, interpolating to finer grids does not introduce new oscillatory structure. This makes the model’s predictions smooth across resolutions, which is essential for the super-resolution evaluation in chapter 5.

6.1.3 Why No Self-Attention Among Queries

An alternative to the cross-attention design would use self-attention among the query points: each query (t_i, x_j) attends to all other queries at the same timestep, allowing the model to share information across spatial locations. This is the standard approach in vision transformers [14], where spatial tokens attend to each other.

The issue is again resolution dependence. Self-attention among queries creates a coupling between query positions: the representation at x_j depends on the set of all other positions $\{x_1, \dots, x_{n_x}\} \setminus \{x_j\}$. When the spatial resolution changes from n_x to n'_x , the attention pattern changes: each query sees a different number of neighbors at different relative positions. The model must implicitly learn how to handle varying neighbor counts and spacings, which is difficult to generalize from a single training resolution.

In the cross-attention design, each query (t, x) attends independently to the K segment embeddings. Since the segment count is determined by the IC, not the query grid, the attention operates over a resolution-independent key-value set. Two grids with $n_x = 50$ and $n_x = 200$ produce query embeddings at different positions, but each query attends to the same segments with the same bias values (because the bias depends on (t, x) and the IC, not on other queries). This makes the model’s behavior invariant to the evaluation resolution.

6.1.4 Implicit $1/t$ Scaling vs. Explicit Collision-Time Damping

The physics-informed bias is derived from local Riemann problems at each interface and is therefore exact only before waves from neighboring interfaces interact. After two waves collide, the initial Riemann structure breaks down and the bias may become misleading.

One approach to handle this is explicit collision-time damping: compute the time $t_{\text{coll}}(k)$ at which neighboring waves reach segment k , and multiply the bias by a sigmoid factor $\sigma(\beta \cdot (t_{\text{coll}} - t))$ that fades the bias after the estimated collision. This requires computing collision times analytically from the IC, which is straightforward for simple ICs but becomes fragile for ICs with many narrow segments where multiple collisions interact.

The similarity-variable formulation (section 4.4.3) provides an implicit alternative. Because the penalty is computed in $\xi = (x - x_d)/(t + \varepsilon)$ coordinates, its magnitude in the attention logits scales as $1/t$: at early times the bias dominates the softmax and enforces the physics prior strongly, while at later times it shrinks and the learned $QK^\top$ scores can override it. This is desirable because the Riemann-based bias is most accurate precisely at early times (before collisions) and least accurate at later times (after collisions), so the prior strength naturally tracks its own reliability.

The default WaveNO uses the implicit $1/t$ scaling without explicit damping. This avoids the need to compute collision times, removes the learnable sharpness parameter β , and lets the model learn when and how to override the prior through the attention scores rather than through a hard sigmoid cutoff. Combined with FiLM time conditioning (section 4.3), which evolves segment embeddings over time, the model has sufficient capacity to represent post-collision dynamics without an explicit damping mechanism.

6.2 Interpreting the Experimental Results

This section revisits three aspects of the experimental results that warrant deeper analysis: the comparison between WaveNO and the Godunov numerical solver, the confound in resolution generalization for equations with numerical ground truth, and the degradation of the physics-informed bias on complex or out-of-distribution initial conditions.

6.2.1 Outperforming a Numerical Solver

On LWR, WaveNO-Full achieves a lower MSE than the first-order Godunov scheme evaluated on the same 50×250 grid (table 5.1). This result requires careful interpretation, as the two methods operate under fundamentally different assumptions.

The Godunov scheme is a grid-bound method: it advances the solution one time step at a time using local Riemann solves at cell interfaces, and its accuracy is limited by the grid spacing. On a coarse grid, numerical diffusion smears shocks over several cells, producing an error that is intrinsic to the discretization and cannot be reduced without refining the mesh. Crucially, Godunov uses no information beyond the initial condition and the PDE: it requires no training data and provides convergence guarantees as the mesh is refined.

WaveNO, by contrast, is a data-driven method trained on 4000 samples whose ground truth comes from the exact Lax–Hopf formula. It has effectively seen the sharp, diffusion-free solution during training and has learned to reproduce it. The comparison is therefore not between two solvers of equal information: WaveNO has access to exact solutions at training time, while Godunov operates from first principles alone. That a learned model can outperform a classical solver on a fixed grid is not surprising in this light; the more meaningful observation is that WaveNO can internalize the structure of the exact solution and generalize it to unseen initial conditions, rather than memorizing individual samples.

6.2.2 Resolution Generalization and Ground-Truth Quality

The resolution generalization results in section 5.4.1 reveal a striking asymmetry between the two equations. On LWR, where reference solutions are computed with the exact Lax–Hopf formula, the WaveNO variants maintain nearly constant MSE from $1\times$ to $5\times$ resolution, confirming that the architecture generalizes to unseen grid spacings. On ARZ, however, *all* models, including WaveNO, degrade significantly at higher resolutions.

This divergence is not a failure of resolution invariance but a confound in the evaluation protocol. The ARZ reference solutions are obtained from a finite volume solver (HLL flux, WENO-5 reconstruction) whose accuracy improves with the grid: at higher resolution, the solver produces sharper shocks and thinner contact discontinuities, yielding a target that is qualitatively different from the training-resolution ground truth. The models were trained against solutions with the numerical diffusion characteristic of the base 50×250 grid; at finer resolutions, they are compared against a more demanding target that contains features they have never seen. The LWR evaluation does not suffer from this confound because the Lax–Hopf formula is exact at every resolution.

This observation has a practical implication: resolution generalization experiments for systems with numerical ground truth should be interpreted with caution, and ideally the comparison should control for ground-truth quality, for instance by evaluating all resolutions against a common high-resolution reference.

6.2.3 The Bias Override Problem

The IC complexity analysis in section 5.4.2 reveals that the physics-informed bias can become counterproductive for complex or out-of-distribution initial conditions. Understanding why overriding an incorrect prior is harder than learning without one sheds light on a general tension in physics-informed architectures.

The biased attention logits are $QK^\top / \sqrt{d} + \text{bias}$ (4.13). When the bias is correct (inside the influence zone, $\text{bias} = 0$), it does not interfere with the learned scores. When it is incorrect, for instance after wave collisions invalidate the local Riemann structure, a large negative bias suppresses a segment that is in fact relevant. To recover the correct attention, the learned $QK^\top$ scores must produce a sufficiently large positive value to cancel the negative bias and restore a meaningful softmax weight.

This cancellation is an extrapolation task. During training, the model observes initial conditions with a bounded range of segment counts and learns $QK^\top$ patterns that override the bias for those configurations. At test time, an out-of-distribution IC with more segments produces wave interactions earlier and at more locations simultaneously, creating bias errors that are both larger in magnitude and more spatially widespread than anything seen during training. The model has not learned the override pattern for this regime, and the residual incorrect bias degrades the prediction.

WaveNO-Bare does not face this failure mode: without a bias to cancel, the learned scores are free to attend to any segment regardless of how many interactions have occurred. This explains the crossover observed in fig. 5.7, where Bare outperforms the biased variants on the most complex ICs.

Several directions could mitigate this trade-off. A learnable scalar gate $\alpha \in [0, 1]$ multiplying the bias would let the model attenuate the prior globally when it detects that the IC is complex, at the cost of an additional learned parameter per layer. Alternatively, curriculum training that progressively increases the segment count during training could expose the model to the override regime more gradually, improving generalization to higher complexity. Finally, the autoregressive approach discussed in chapter 7 would sidestep the problem entirely by recomputing the bias from the current predicted state at each time step, keeping the Riemann-based prior accurate throughout the time horizon.

Conclusion

7.1 Summary

This thesis introduced WaveNO, a transformer-based neural operator for hyperbolic conservation laws that embeds the physics of wave propagation directly into its architecture. The key design choices are: (i) operating on the exact piecewise constant description of the initial condition rather than a discretized grid, (ii) augmenting cross-attention with a physics-informed bias derived from local Riemann problems, and (iii) evolving segment representations over time via FiLM conditioning. (ii) adds no complexity to the model while (iii) is of negligible cost.

On the LWR traffic flow equation, WaveNO reduces test MSE by 63% over FNO and outperforms a first-order Godunov solver evaluated on the same grid. On the ARZ system, it achieves a 30% improvement over FNO. The ablation study shows that the physics-informed bias contributes most at early time steps, when the local Riemann structure is still intact, and that FiLM conditioning best captures post-interaction dynamics.

7.2 Limitations

The physics-informed bias is computed from local Riemann problems and is exact only before neighboring waves interact. For initial conditions with many segments, waves collide early and the bias becomes inaccurate for most of the time horizon. The model must then learn to override the bias through the attention scores, which it cannot do reliably for out-of-distribution segment counts not seen during training. In this regime, WaveNO-Bare, which has no bias to override, can be preferable.

Finally, the restriction to piecewise constant initial conditions, while natural for Riemann-based analysis and common in traffic flow modeling, limits

applicability to problems with smooth or more general initial data.

7.3 Future Work

Two directions follow naturally from the current limitations.

Autoregressive prediction. The current architecture predicts the full space-time solution in a single forward pass. An autoregressive variant that predicts one or a few time steps at a time could use the physics-informed bias more effectively: at each step, the bias would be recomputed from the current state rather than the initial condition, keeping the Riemann-based prior accurate even after wave interactions. This would address the bias degradation observed at late times and for complex initial conditions.

Beyond piecewise constant initial conditions. Piecewise constant functions are a natural starting point because they admit exact Riemann solutions at each interface, but real-world applications often involve smooth or piecewise linear initial data. There may be ways to extend the input representation with corresponding adaptations to the characteristic feature computation that would broaden the applicability of WaveNO while preserving the mesh-free input representation.

Bibliography

- [1] Benedikt Alkin, Andreas Fürst, Simon Schmid, Lukas Gruber, Markus Holzleitner, and Johannes Brandstetter. Universal physics transformers: A framework for efficiently scaling neural operators. *arXiv preprint arXiv:2402.12365*, 2024.
- [2] A. Aw and M. Rascle. Resurrection of “second order” models of traffic flow. *SIAM Journal on Applied Mathematics*, 60(3):916–938, 2000.
- [3] Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E. Hinton. Layer normalization. *arXiv preprint arXiv:1607.06450*, 2016.
- [4] Marsha J. Berger and Phillip Colella. Local adaptive mesh refinement for shock hydrodynamics. *Journal of Computational Physics*, 82(1):64–84, 1989.
- [5] Edoardo Calvello, Nikola B. Kovachki, Matthew E. Levine, and Andrew M. Stuart. Continuum attention for neural operators. *arXiv preprint arXiv:2406.06486*, 2024.
- [6] Shuhao Cao. Choose a transformer: Fourier or Galerkin. In *Advances in Neural Information Processing Systems*, volume 34, 2021.
- [7] Tianping Chen and Hong Chen. Universal approximation to nonlinear operators by neural networks with arbitrary activation functions and its application to dynamical systems. *IEEE Transactions on Neural Networks*, 6(4):911–917, 1995.
- [8] Ta-Chung Chi, Ting-Han Fan, Alexander Rudnicky, and Peter Ramadge. Dissecting transformer length extrapolation via the lens of receptive field analysis. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13522–13537. Association for Computational Linguistics, 2023.

- [9] Bernardo Cockburn and Chi-Wang Shu. TVB Runge–Kutta local projection discontinuous Galerkin finite element method for scalar conservation laws II: General framework. *Mathematics of Computation*, 52(186):411–435, 1989.
- [10] Emilio Jose Rocha Coutinho, Marcelo Dall’Aqua, Levi McClenny, Ming Zhong, Ulisses Braga-Neto, and Eduardo Gildin. Physics-informed neural networks with adaptive localized artificial viscosity. *Journal of Computational Physics*, 489:112265, 2023.
- [11] Constantine M. Dafermos. *Hyperbolic Conservation Laws in Continuum Physics*, volume 325 of *Grundlehren der mathematischen Wissenschaften*. Springer, Berlin, Heidelberg, 4th edition, 2016.
- [12] Tim De Ryck and Siddhartha Mishra. Generic bounds on the approximation error for physics-informed (and) operator learning. In *Advances in Neural Information Processing Systems*, volume 35, 2022.
- [13] Tim De Ryck, Siddhartha Mishra, and Roberto Molinaro. wPINNs: Weak physics informed neural networks for approximating entropy solutions of hyperbolic conservation laws. *SIAM Journal on Numerical Analysis*, 62(2):811–841, 2024.
- [14] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2021.
- [15] Lawrence C. Evans. *Partial Differential Equations*, volume 19 of *Graduate Studies in Mathematics*. American Mathematical Society, Providence, RI, 2nd edition, 2010.
- [16] Edwige Godlewski and Pierre-Arnaud Raviart. *Numerical Approximation of Hyperbolic Systems of Conservation Laws*, volume 118 of *Applied Mathematical Sciences*. Springer, New York, 1996.
- [17] Sergei K. Godunov. A difference method for numerical calculation of discontinuous solutions of the equations of hydrodynamics. *Matematicheskii Sbornik*, 47(89(3)):271–306, 1959.
- [18] Bruce D. Greenshields. A study of traffic capacity. *Highway Research Board Proceedings*, 14:448–477, 1935.

-
- [19] Zhongkai Hao, Chang Su, Songming Liu, Julius Berner, Chengyang Ying, Hang Su, Anima Anandkumar, Jian Song, and Jun Zhu. DPOT: Auto-regressive denoising operator transformer for large-scale PDE pre-training. *arXiv preprint arXiv:2403.03542*, 2024.
 - [20] Zhongkai Hao, Zhengyi Wang, Hang Su, Chengyang Ying, Yinpeng Dong, Songming Liu, Ze Cheng, Jian Song, and Jun Zhu. GNOT: A general neural operator transformer for operator learning. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 12556–12569. PMLR, 2023.
 - [21] Ami Harten, Björn Engquist, Stanley Osher, and Sukumar R. Chakravarthy. Uniformly high order accurate essentially non-oscillatory schemes, III. *Journal of Computational Physics*, 71(2):231–303, 1987.
 - [22] Jan S. Hesthaven and Tim Warburton. *Nodal Discontinuous Galerkin Methods: Algorithms, Analysis, and Applications*, volume 54 of *Texts in Applied Mathematics*. Springer, New York, 2008.
 - [23] Jianing Huang, Kaixuan Zhang, Youjia Wu, and Ze Cheng. Operator learning with domain decomposition for geometry generalization in PDE solving. *arXiv preprint arXiv:2504.00510*, 2025.
 - [24] Pierre-Henri Hugoniot. Mémoire sur la propagation du mouvement dans les corps et spécialement dans les gaz parfaits. *Journal de l'École Polytechnique*, 58:1–125, 1889.
 - [25] Ameya D. Jagtap and George Em Karniadakis. Extended physics-informed neural networks (XPINNs): A generalized space-time domain decomposition based deep learning framework for nonlinear partial differential equations. *Communications in Computational Physics*, 28(5):2002–2041, 2020.
 - [26] Ameya D. Jagtap, Ehsan Kharazmi, and George Em Karniadakis. Conservative physics-informed neural networks on discrete domains for conservation laws: Applications to forward and inverse problems. *Computer Methods in Applied Mechanics and Engineering*, 365:113028, 2020.
 - [27] Pengzhan Jin, Shuai Meng, and Lu Lu. MIONet: Learning multiple-input operators via tensor product. *SIAM Journal on Scientific Computing*, 44(6):A3490–A3514, 2022.
 - [28] George Em Karniadakis, Ioannis G. Kevrekidis, Lu Lu, Paris Perdikaris, Sifan Wang, and Liu Yang. Physics-informed machine learning. *Nature Reviews Physics*, 3(6):422–440, 2021.

- [29] Ehsan Kharazmi, Zhongqiang Zhang, and George Em Karniadakis. hp-VPINNs: Variational physics-informed neural networks with domain decomposition. *Computer Methods in Applied Mechanics and Engineering*, 374:113547, 2021.
- [30] Georgios Kissas, Jacob H. Seidman, Leonardo Ferreira Guilhoto, Victor M. Preciado, George J. Pappas, and Paris Perdikaris. Learning operators with coupled attention. *Journal of Machine Learning Research*, 23(215):1–63, 2022.
- [31] Nikola B. Kovachki, Zongyi Li, Burigede Liu, Kamyar Azizzadenesheli, Kaushik Bhattacharya, Andrew M. Stuart, and Anima Anandkumar. Neural operator: Learning maps between function spaces with applications to PDEs. *Journal of Machine Learning Research*, 24(89):1–97, 2023.
- [32] Aditi S. Krishnapriyan, Amir Gholami, Shandian Zhe, Robert M. Kirby, and Michael W. Mahoney. Characterizing possible failure modes in physics-informed neural networks. In *Advances in Neural Information Processing Systems*, volume 34, pages 26548–26560, 2021.
- [33] Samuel Lanthaler, Siddhartha Mishra, and George Em Karniadakis. Error estimates for DeepONets: A deep learning framework in infinite dimensions. *Transactions of Mathematics and Its Applications*, 6(1):tnac001, 2022.
- [34] Samuel Lanthaler, Roberto Molinaro, Patrik Hadorn, and Siddhartha Mishra. Nonlinear reconstruction for operator learning of PDEs with discontinuities. In *International Conference on Learning Representations*, 2023.
- [35] Peter D. Lax. *Hyperbolic Systems of Conservation Laws and the Mathematical Theory of Shock Waves*. Number 11 in CBMS-NSF Regional Conference Series in Applied Mathematics. Society for Industrial and Applied Mathematics, Philadelphia, 1973.
- [36] Randall J. LeVeque. *Finite Volume Methods for Hyperbolic Problems*. Number 31 in Cambridge Texts in Applied Mathematics. Cambridge University Press, 2002.
- [37] Zijie Li, Kazem Meidani, and Amir Barati Farimani. Transformer for partial differential equations’ operator learning. *Transactions on Machine Learning Research*, 2023.
- [38] Zijie Li, Dule Shu, and Amir Barati Farimani. Scalable transformer for PDE surrogate modeling. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

-
- [39] Zongyi Li, Daniel Zhengyu Huang, Burigede Liu, and Anima Anandkumar. Fourier neural operator with learned deformations for PDEs on general geometries. *Journal of Machine Learning Research*, 24(388):1–26, 2023.
- [40] Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Neural operator: Graph kernel network for partial differential equations. *arXiv preprint arXiv:2003.03485*, 2020.
- [41] Zongyi Li, Nikola B. Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations*, 2021.
- [42] Zongyi Li, Hongkai Zheng, Nikola Kovachki, David Jin, Haoxuan Chen, Burigede Liu, Kamyar Azizzadenesheli, and Anima Anandkumar. Physics-informed neural operator for learning partial differential equations. *ACM/IMS Journal of Data Science*, 1(3):1–27, 2024.
- [43] Nathan Lichtlé, Alexi Canesse, Zhe Fu, Hossein Nick Zinat Martin, Maria Laura Delle Monache, and Alexandre M. Bayen. (u)NfV: Supervised and unsupervised neural finite volume methods for solving hyperbolic PDEs. *arXiv preprint arXiv:2505.23702*, 2025.
- [44] M. J. Lighthill and G. B. Whitham. On kinematic waves II. A theory of traffic flow on long crowded roads. *Proceedings of the Royal Society of London. Series A, Mathematical and Physical Sciences*, 229(1178):317–345, 1955.
- [45] Xu-Dong Liu, Stanley Osher, and Tony Chan. Weighted essentially non-oscillatory schemes. *Journal of Computational Physics*, 115(1):200–212, 1994.
- [46] Emmanuel Lorin and Arian Novruzli. Non-diffusive neural network method for hyperbolic conservation laws. *arXiv preprint arXiv:2405.15559*, 2024.
- [47] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *7th International Conference on Learning Representations, ICLR*. OpenReview.net, 2019.
- [48] Lu Lu, Pengzhan Jin, Guofei Pang, Zhongqiang Zhang, and George Em Karniadakis. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. *Nature Machine Intelligence*, 3(3):218–229, 2021.

-
- [49] Lu Lu, Xuhui Meng, Shengze Cai, Zhiping Mao, Somdatta Goswami, Zhongqiang Zhang, and George Em Karniadakis. A comprehensive and fair comparison of two neural operators (with practical extensions) based on FAIR data. *Computer Methods in Applied Mechanics and Engineering*, 393:114778, 2022.
 - [50] Michael McCabe, Bruno Régaldo-Saint Blancard, Liam Holden Parker, Ruben Ohana, Miles Cranmer, Alberto Bietti, Michael Eickenberg, Siavash Golkar, Geraud Krawezik, Francois Lanusse, Mariel Pet-tee, Tiberiu Tesileanu, Kyunghyun Cho, and Shirley Ho. Multiple physics pretraining for physical surrogate models. *arXiv preprint arXiv:2310.02994*, 2023.
 - [51] Stanley Osher. Riemann solvers, the entropy condition, and difference approximations. *SIAM Journal on Numerical Analysis*, 21(2):217–235, 1984.
 - [52] Oded Ovadia, Adar Kahana, Panos Stinis, Eli Turkel, Dan Givoli, and George Em Karniadakis. ViTO: Vision transformer-operator. *Computer Methods in Applied Mechanics and Engineering*, 428:117109, 2024.
 - [53] Ethan Perez, Florian Strub, Harm de Vries, Vincent Dumoulin, and Aaron Courville. FiLM: Visual reasoning with a general conditioning layer. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, 2018.
 - [54] Philipp Petersen and Felix Voigtländer. Optimal approximation of piecewise smooth functions using deep ReLU neural networks. *Neural Networks*, 108:296–330, 2018.
 - [55] Ofir Press, Noah A. Smith, and Mike Lewis. Train short, test long: Attention with linear biases enables input length generalization. In *International Conference on Learning Representations*, 2022.
 - [56] Nasim Rahaman, Aristide Baratin, Devansh Arpit, Felix Draxler, Min Lin, Fred A. Hamprecht, Yoshua Bengio, and Aaron Courville. On the spectral bias of neural networks. In *International Conference on Machine Learning*, pages 5301–5310, 2019.
 - [57] Md Ashiqur Rahman, Robert Joseph George, Mogab Elleithy, Daniel Leibovici, Zongyi Li, Boris Bonev, Colin White, Julius Berner, Raymond A. Yeh, Jean Kossaifi, Kamyar Azizzadenesheli, and Anima Anandkumar. Pretraining codomain attention neural operators for solving multi-physics PDEs. *arXiv preprint arXiv:2403.12553*, 2024.
 - [58] Md Ashiqur Rahman, Zachary E. Ross, and Kamyar Azizzadenesheli. U-NO: U-shaped neural operators. *Transactions on Machine Learning Research*, 2023.

- [59] Maziar Raissi, Paris Perdikaris, and George Em Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.
- [60] William John Macquorn Rankine. On the thermodynamic theory of waves of finite longitudinal disturbance. *Philosophical Transactions of the Royal Society of London*, 160:277–288, 1870.
- [61] Paul I. Richards. Shock waves on the highway. *Operations Research*, 4(1):42–51, 1956.
- [62] Ruben Rodriguez-Torrado, Pablo Ruiz, Luis Cueto-Felgueroso, Michael Cerny Green, Sébastien Matringe, and Julian Togelius. Physics-informed attention-based neural network for hyperbolic partial differential equations: application to the Buckley–Leverett problem. *Scientific Reports*, 12:7557, 2022.
- [63] P. L. Roe. Approximate Riemann solvers, parameter vectors, and difference schemes. *Journal of Computational Physics*, 43(2):357–372, 1981.
- [64] Chi-Wang Shu. High order weighted essentially nonoscillatory schemes for convection dominated problems. *SIAM Review*, 51(1):82–126, 2009.
- [65] Matthew Tancik, Pratul P. Srinivasan, Ben Mildenhall, Sara Fridovich-Keil, Nithin Raghavan, Utkarsh Singhal, Ravi Ramamoorthi, Jonathan T. Barron, and Ren Ng. Fourier features let networks learn high frequency functions in low dimensional domains. In *Advances in Neural Information Processing Systems*, volume 33, pages 7537–7547, 2020.
- [66] Bilal Thonnam Thodi, Sai Venkata Ramana Ambadipudi, and Saif Eddin Jabari. Learning-based solutions to nonlinear hyperbolic PDEs: Empirical insights on generalization errors. *arXiv preprint arXiv:2302.08144*, 2023.
- [67] Eleuterio F. Toro. *Riemann Solvers and Numerical Methods for Fluid Dynamics: A Practical Introduction*. Springer, Berlin, Heidelberg, 3rd edition, 2009.
- [68] Alasdair Tran, Alexander Mathews, Lexing Xie, and Cheng Soon Ong. Factorized Fourier neural operators. In *International Conference on Learning Representations*, 2023.
- [69] Tapas Tripura and Souvik Chakraborty. Wavelet neural operator for solving parametric partial differential equations in computational mechanics problems. *Computer Methods in Applied Mechanics and Engineering*, 404:115783, 2023.

- [70] Bram van Leer. Towards the ultimate conservative difference scheme. V. A second-order sequel to Godunov's method. *Journal of Computational Physics*, 32(1):101–136, 1979.
- [71] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, pages 6000–6010, 2017.
- [72] Christopher Wang and Alex Townsend. Operator learning for hyperbolic partial differential equations. *arXiv preprint arXiv:2312.17489*, 2023.
- [73] Sifan Wang, Shyam Sankaran, Hanwen Wang, and Paris Perdikaris. An expert's guide to training physics-informed neural networks. *arXiv preprint arXiv:2308.08468*, 2023.
- [74] Sifan Wang, Jacob H. Seidman, Shyam Sankaran, Hanwen Wang, George J. Pappas, and Paris Perdikaris. CViT: Continuous vision transformer for operator learning. *arXiv preprint arXiv:2405.13998*, 2024.
- [75] Sifan Wang, Xinling Yu, and Paris Perdikaris. When and why PINNs fail to train: A neural tangent kernel perspective. *Journal of Computational Physics*, 449:110768, 2022.
- [76] Haixu Wu, Minghao Guo, Yuezhou Ma, Yuanxu Sun, Jianmin Wang, Wojciech Matusik, and Mingsheng Long. FlashBias: Fast computation of attention with bias. In *Advances in Neural Information Processing Systems*, 2025.
- [77] Haixu Wu, Huakun Luo, Haowen Wang, Jianmin Wang, and Mingsheng Long. Transolver: A fast transformer solver for PDEs on general geometries. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 53681–53705. PMLR, 2024.
- [78] H. M. Zhang. A non-equilibrium traffic model devoid of gas-like behavior. *Transportation Research Part B: Methodological*, 36(3):275–290, 2002.

Appendix

Experiment Details

This section provides the full training configuration and model architecture hyperparameters for the experiments reported in chapter 5.

Training Hyperparameters

All models are trained with the same optimizer and regularization settings, listed in table 1. We use AdamW [47] with decoupled weight decay rather than the original Adam formulation. The learning rate is reduced on plateau using the ReduceLROnPlateau scheduler, and training is stopped early if the validation loss does not improve for 15 consecutive epochs.

Per-Equation Configuration

Table 2 details the equation-specific settings. For each equation, 5000 samples are generated in total: 4000 for training, 500 for validation, and 500 for testing. The number of IC segments at training time is drawn uniformly from $\{2, \dots, K_{\text{train}}\}$, where K_{train} differs per equation. At test time, models are additionally evaluated on higher segment counts (up to K_{test}) to assess generalization to more complex initial conditions.

Model Architectures

Table 3 lists the architectural hyperparameters for each model. All models adapt their output channels to the equation: 1 for LWR (density), 2 for ARZ (density and velocity).

The FNO uses 2 Fourier layers with 32 hidden channels, retaining 32 modes along the time axis and 16 along the spatial axis. DeepONet uses a branch network (4 layers, 128 hidden units) to encode the discretized IC and a

Table 1: Training hyperparameters shared across all models and equations.

Hyperparameter	Value
Optimizer	AdamW
Learning rate	10^{-3}
Weight decay	10^{-4}
Batch size	8
Epochs	100
Loss	MSE
LR scheduler	ReduceLROnPlateau
Factor	0.5
Patience	5 epochs
Threshold	0.01
Early stopping patience	15 epochs
Gradient clipping	max norm = 1.0

Table 2: Per-equation data and evaluation configuration.

	LWR	ARZ
Samples (train / val / test)	4000 / 500 / 500	4000 / 500 / 500
IC segments (train), K_{train}	8	6
IC segments (test), K_{test}	16	12
Ground truth solver	Lax–Hopf (exact)	FV (HLL + WENO-5 + SSP-RK3)

Table 3: Model architecture hyperparameters.

	FNO	DeepONet	WaveNO variants
Layers	2 Fourier	4 branch + 4 trunk	2 self-attn + 2 cross-attn
Hidden dim	32	128	64
Fourier modes	$32_t \times 16_x$	—	—
Latent / heads	—	64 (dot-product)	4 heads
Segment MLP	—	—	2 layers
Dropout	—	—	0.05

trunk network (4 layers, same width) to encode the (t, x) query coordinates, combining them via a dot product in a 64-dimensional latent space.

All WaveNO variants share the same core architecture: a 2-layer segment encoder MLP lifts each IC segment to 64-dimensional embeddings, followed by 2 self-attention layers for segment interaction and 2 cross-attention layers where (t, x) queries attend to the segment embeddings. The variants differ only in whether the physics-informed attention bias and FiLM time conditioning are enabled, as described in section 5.1.2.

Additional Experimental Results

Full Solution Comparison

Figures 1 and 2 show the full space-time solution grids for a representative test sample on each equation. Each row corresponds to a model, and the three columns show the ground truth, the model prediction, and the pointwise squared error. The error colorscale is shared across all rows within each figure, making the relative accuracy directly comparable.

On LWR (fig. 1), FNO produces visible oscillations near the shock front and spreads the error across a broad region of the domain. DeepONet captures the general structure but smears the shock, resulting in a diffuse error band. The Godunov scheme shows numerical diffusion along the shock trajectory, as expected for a first-order finite volume method. All three WaveNO variants concentrate their error tightly along the shock front, with WaveNO-Full achieving the smallest error magnitude.

On ARZ (fig. 2), the pattern is consistent: FNO and DeepONet show broad error regions, particularly along the contact discontinuity and shock trajectories, while the WaveNO variants produce sharply localized errors. The progression from WaveNO-Bare to WaveNO-Full shows a visible reduction in error intensity.

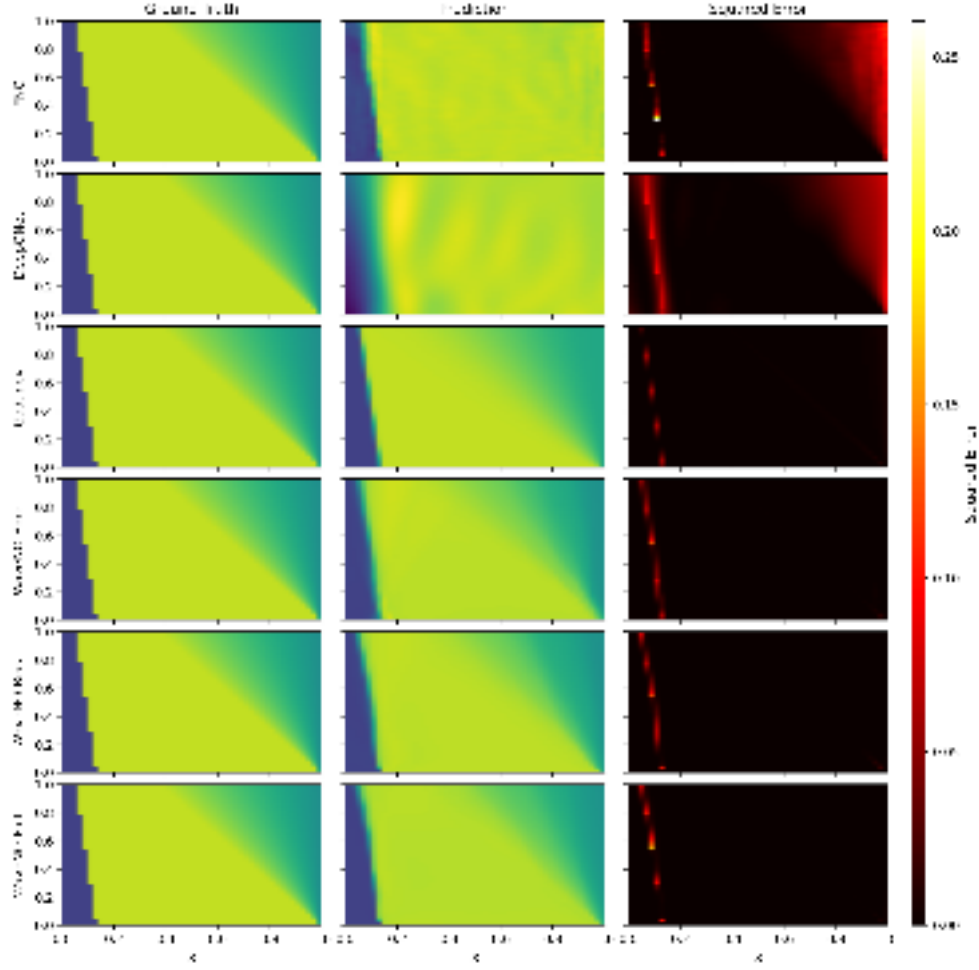


Figure 1: LWR model comparison on a representative test sample. Rows: FNO, DeepONet, Godunov, WaveNO-Bare, WaveNO-Bias, WaveNO-Full. Columns: ground truth, prediction, pointwise squared error (shared colorscale). WaveNO variants concentrate error at the shock front, while baselines spread error across the domain.

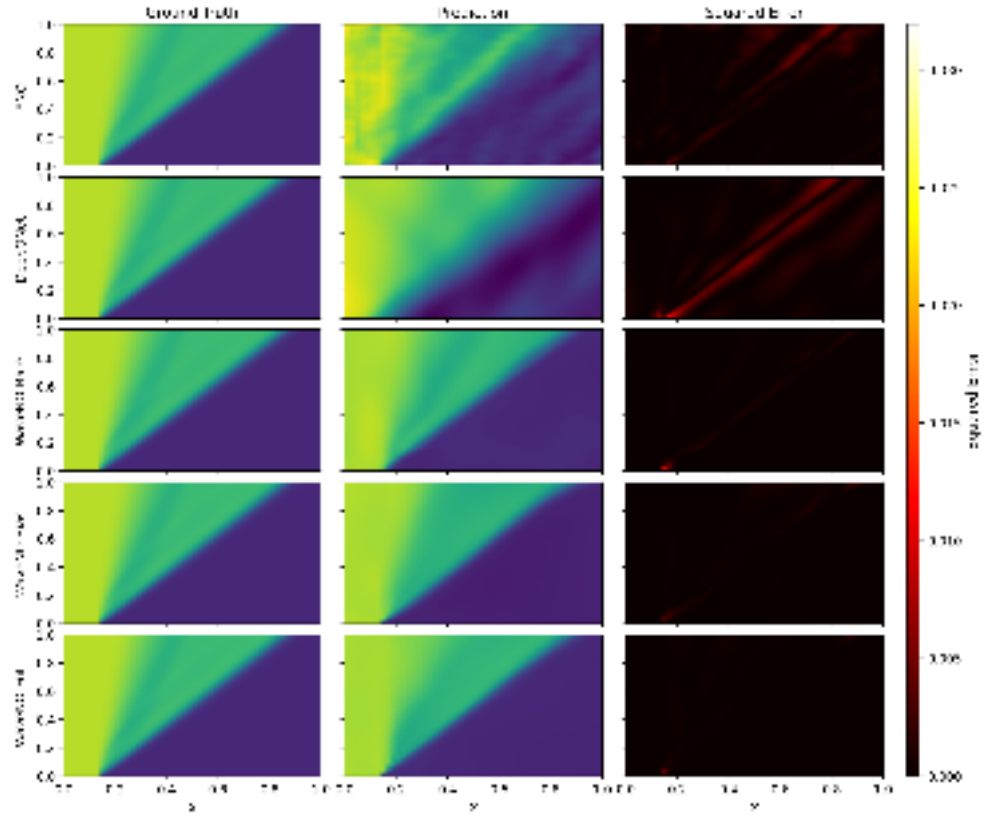


Figure 2: ARZ density channel comparison on a representative test sample. Rows: FNO, DeepONet, WaveNO-Bare, WaveNO-Bias, WaveNO-Full. Columns: ground truth, prediction, pointwise squared error (shared colorscale). The WaveNO variants track the shock and contact discontinuity more accurately than the baselines.

Declaration of originality

The signed declaration of originality is a component of every written paper or thesis authored during the course of studies. In consultation with the supervisor, one of the following three options must be selected:

- ☐ I confirm that I authored the work in question independently and in my own words, i.e. that no one helped me to author it. Suggestions from the supervisor regarding language and content are excepted. I used no generative artificial intelligence technologies¹.
- ☐ I confirm that I authored the work in question independently and in my own words, i.e. that no one helped me to author it. Suggestions from the supervisor regarding language and content are excepted. I used and cited generative artificial intelligence technologies².
- ☒ I confirm that I authored the work in question independently and in my own words, i.e. that no one helped me to author it. Suggestions from the supervisor regarding language and content are excepted. I used generative artificial intelligence technologies³. In consultation with the supervisor, I did not cite them.

Title of paper or thesis:

WaveNO: Solving PDEs using Physics-Informed Attention Bias

Authored by:

If the work was compiled in a group, the names of all authors are required.

Last name(s):

Chami

First name(s):

Mamoun

With my signature I confirm the following:

- I have adhered to the rules set out in the Citation Guide.
- I have documented all methods, data and processes truthfully and fully.
- I have mentioned all persons who were significant facilitators of the work.

I am aware that the work may be screened electronically for originality.

Place, date

Berkeley, 01/04/2026

Signature(s)

Mamoun Chami

If the work was compiled in a group, the names of all authors are required. Through their signatures they vouch jointly for the entire content of the written work.

¹ E.g. ChatGPT, DALL E 2, Google Bard

² E.g. ChatGPT, DALL E 2, Google Bard

³ E.g. ChatGPT, DALL E 2, Google Bard